

SPARSE WAVELET ESTIMATION OF ADDITIVE FUNCTIONAL COEFFICIENTS

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Supplementary Material

This supplement contains the proofs of the results stated in the paper and the auxiliary lemmas they use. Section S1 fixes the notation and the objects. Section S2 proves the identification results of Section 2 of the paper, Section S3 the approximation lemma, Section S4 the bounds for the Gram matrix of the product design, Section S5 the oracle inequality, and Section S6 the rates, and Section S7 the recovery of the block structure by thresholding and the risk of the thresholded estimator, and Section S8 gives the theory of the coordinatewise lasso, the option of the software that replaces the penalty of equation (2.5) by the ℓ_1 norm. References to numbered results without a section prefix are to the paper; results numbered with the prefix S are stated and proved here.

S1 Notation and objects

The model, the basis and the estimator are those of Sections 2.1 to 2.3 of the paper. We write $\mathbf{Z} = [\mathbf{A} \ \mathbf{B}]$ for the design matrix of equation (2.4), with $\mathbf{A} \in \mathbb{R}^{n \times p}$ the unpenalized block and $\mathbf{B} \in \mathbb{R}^{n \times d}$, $d = pqN_J$, the block

of products; $\mathbf{Y}$, f and $\boldsymbol{\varepsilon}$ for the vectors with entries Y_i , $f(\mathbf{X}_i, \mathbf{U}_i)$ and ε_i ; $\widehat{\Sigma} = \mathbf{Z}^\top \mathbf{Z}/n$ and $\Sigma = \mathbb{E}(\mathbf{Z}\mathbf{Z}^\top)$; and, for $a, h \in \mathbb{R}^n$, $\|a\|_n^2 = n^{-1} \sum_i a_i^2$ and $\langle a, h \rangle_n = n^{-1} \sum_i a_i h_i$.

The oracle of the approximation space is $\boldsymbol{\beta}^* = (\mathbf{c}, \boldsymbol{\theta}^*)$, with $\theta_{\ell m, jk}^* = \langle g_{\ell m}, \psi_{jk} \rangle$ for $j < J$, so that $f_J = \mathbf{Z}\boldsymbol{\beta}^*$ is the best approximation of f in the approximation space in the sense of Lemma 2; the deterministic residual is $\mathbf{b} = f - f_J$, the oracle support is $S_0 = \text{supp}(\boldsymbol{\theta}^*)$ and $s_0 = |S_0|$, and $v = \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$. When $\mathbf{A}$ has rank p we write $\Pi_{\mathbf{A}} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$, $M_{\mathbf{A}} = I_n - \Pi_{\mathbf{A}}$, $\widetilde{\mathbf{B}} = M_{\mathbf{A}} \mathbf{B}$, $\widehat{\Sigma}_{\mathbf{B}|\mathbf{A}} = \widetilde{\mathbf{B}}^\top \widetilde{\mathbf{B}}/n$ and $\widetilde{\gamma} = \lambda_{\min}(\widehat{\Sigma}_{\mathbf{B}|\mathbf{A}})$, and finally $\widehat{\sigma}_{\max}^2 = \max_a (\widehat{\Sigma})_{aa}$, the maximum being over the d penalized columns.

For the penalty of equation (2.5), $\mathcal{G}$ is the partition of the d penalized coordinates into the chunks of Section 2.3 of the paper, $|\mathcal{G}|$ the number of chunks, $\boldsymbol{\theta}_G$ and $|G|$ the subvector and the size of a chunk G , $w_G > 0$ its weight and $\varrho = \max_G w_G / \min_G w_G$ the ratio of the weights; $\|\boldsymbol{\theta}\|_{\mathcal{G}, w} = \sum_G w_G \|\boldsymbol{\theta}_G\|_2$, $\|\boldsymbol{\theta}\|_{\mathcal{G}, 1}$ is the same norm with unit weights, and, for $\mathcal{H} \subseteq \mathcal{G}$, $\|\boldsymbol{\theta}_{\mathcal{H}}\|_{\mathcal{G}, w} = \sum_{G \in \mathcal{H}} w_G \|\boldsymbol{\theta}_G\|_2$ and $\mathcal{W}(\mathcal{H}) = \sum_{G \in \mathcal{H}} w_G^2$; $\mathcal{G}_0(\boldsymbol{\theta}) = \{G : \boldsymbol{\theta}_G \neq 0\}$ are the active chunks of $\boldsymbol{\theta}$. With unit weights $\mathcal{W}(\mathcal{G}_0(\boldsymbol{\theta}))$ is the number of active chunks, and with $w_G = |G|^{1/2}$ it is the number of coordinates in them. A comparator is any $\bar{\boldsymbol{\beta}} = (\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}}) \in \mathbb{R}^p \times \mathbb{R}^d$, with deterministic residual $\bar{\mathbf{b}} = f - \mathbf{Z}\bar{\boldsymbol{\beta}}$, active chunks $\bar{\mathcal{G}} = \mathcal{G}_0(\bar{\boldsymbol{\theta}})$, $\bar{\mathcal{W}} = \mathcal{W}(\bar{\mathcal{G}})$ and $\bar{v} = \widehat{\boldsymbol{\theta}} - \bar{\boldsymbol{\theta}}$. For a chunk,

$\mathbf{B}_G$ and $\tilde{\mathbf{B}}_G$ are the columns of $\mathbf{B}$ and $\tilde{\mathbf{B}}$ in G , $\tilde{\Psi}_G = \tilde{\mathbf{B}}_G^\top \tilde{\mathbf{B}}_G / n$ and $\hat{\Sigma}_{GG} = \mathbf{B}_G^\top \mathbf{B}_G / n$. The ideal risk over chunks is $R_G(\boldsymbol{\theta}; \eta) = \sum_G \min(\|\boldsymbol{\theta}_G\|_2^2, \eta)$, for $\eta > 0$; it is nondecreasing in η , and, since $\min(u, \kappa\eta) \leq \kappa \min(u, \eta)$ for $\kappa \geq 1$, $R_G(\boldsymbol{\theta}; \kappa\eta) \leq \kappa R_G(\boldsymbol{\theta}; \eta)$. Finally $\gamma = \kappa_1 c_U$, $\Lambda = \kappa_2 C_U + \gamma/2$ is the constant of equation (3.1), and

$$\mathcal{E}_\Sigma = \{ \|\hat{\Sigma} - \Sigma\|_{\text{op}} < \gamma/2 \};$$

on $\mathcal{E}_\Sigma$, Weyl's inequality and Proposition 2(i) give $\lambda_{\min}(\hat{\Sigma}) > \gamma/2$ and $\lambda_{\max}(\hat{\Sigma}) < \Lambda$, and the matrix Bernstein bound in the proof of Proposition 2(ii), Section S4, gives $\mathbb{P}(\mathcal{E}_\Sigma) \rightarrow 1$ whenever $pq2^J \log(pq2^J)/n \rightarrow 0$.

For a support S and a symmetric matrix A , the compatibility constant is

$$\phi_0^2(S; A) = \min\{|S| \delta^\top A \delta : \|\delta_S\|_1 = 1, \|\delta_{S^c}\|_1 \leq 3\},$$

and the elementary bound $\delta^\top A \delta \geq \lambda_{\min}(A) \|\delta\|_2^2 \geq \lambda_{\min}(A) \|\delta_S\|_1^2 / |S|$ gives

$$\phi_0^2(S; A) \geq \lambda_{\min}(A) \quad \text{for every } S, \quad (\text{S1.1})$$

so that a lower bound on the smallest eigenvalue is a compatibility bound on every support, with the same constant and no cone restriction.

S2 Identification and the periodized basis

Proof of Proposition 1

Levels and components are determined by the coefficients. Let β and $\tilde{\beta}$ be measurable maps from $[0, 1]^q$ to $\mathbb{R}^p$ with $\sum_{\ell} \beta_{\ell}(\mathbf{U}) X_{\ell} = \sum_{\ell} \tilde{\beta}_{\ell}(\mathbf{U}) X_{\ell}$ almost surely, and put $\delta(u) = \beta(u) - \tilde{\beta}(u)$. Then $\{\delta(\mathbf{U})^{\top} \mathbf{X}\}^2 = 0$ almost surely, and since $\delta(\mathbf{U})$ is measurable with respect to $\mathbf{U}$ and $E \|\mathbf{X}\|_2^2 < \infty$,

$$0 = E[\{\delta(\mathbf{U})^{\top} \mathbf{X}\}^2 \mid \mathbf{U}] = \delta(\mathbf{U})^{\top} E(\mathbf{X}\mathbf{X}^{\top} \mid \mathbf{U}) \delta(\mathbf{U}) \quad \text{almost surely.}$$

The conditional second moment matrix is positive definite almost surely, and a positive definite quadratic form vanishes only at the origin, so $\delta(\mathbf{U}) = 0$ almost surely.

Levels and components are determined within a coefficient. Because the density of $\mathbf{U}$ is positive Lebesgue almost everywhere, a set is null for the law of $\mathbf{U}$ if and only if it is Lebesgue null, so the identity $c_{\ell} + \sum_m g_{\ell m}(u_m) = \tilde{c}_{\ell} + \sum_m \tilde{g}_{\ell m}(u_m)$ holds Lebesgue almost everywhere on $[0, 1]^q$. Write $\Delta = c_{\ell} - \tilde{c}_{\ell}$ and $\delta_m = g_{\ell m} - \tilde{g}_{\ell m} \in L_2[0, 1] \subset L_1[0, 1]$, so that $\Delta + \sum_m \delta_m(u_m) = 0$ almost everywhere. Each summand is integrable on the cube by Tonelli's theorem. Integrating in all coordinates and using $\int_0^1 \delta_m = 0$, which holds because both parametrizations satisfy (2.2), gives $\Delta = 0$; fixing m and integrating in the remaining coordinates gives $\delta_m(u_m) = 0$ for almost every u_m . $\square$

Remark S2.1. If the centering $\int_0^1 g_{\ell m} = 0$ is replaced by $E\{g_{\ell m}(U_m)\} = 0$, the same proof applies under the two sided density bound of Assumption 3, since then $L_2[0, 1]$ and $L_2(P_{U_m})$ coincide as sets: the step that uses Fubini's theorem yields δ_m constant almost everywhere, and the constant is its mean, which is zero. The two parametrizations are related by $c_\ell \mapsto c_\ell + \sum_m E\{g_{\ell m}(U_m)\}$, as stated in Section 2.1 of the paper. The difference is not cosmetic for the estimator: the design of Section 2.3, which omits the constant scaling function, estimates the version centered with respect to Lebesgue measure.

Proof of Lemma 1

The basis. The periodization of the scaling function is $\phi_{00}^{\text{per}}(u) = \sum_{\iota} \phi(u + \iota)$, a finite sum at each u because the support is compact, and it equals the constant function one, since the integer translates of an orthonormal scaling function with $\int \phi = 1$ form a partition of unity ([Mallat, 2009, Section 7.5.1](#)). For a wavelet, exchanging the finite sum with the integral, $\int_0^1 \psi_{jk} = \int_{\mathbb{R}} 2^{j/2} \psi(2^j y - k) dy = 2^{-j/2} \int_{\mathbb{R}} \psi = 0$ by the vanishing moment of order zero. That $\{\mathbf{1}\} \cup \{\psi_{jk}\}$ is an orthonormal basis of $L_2[0, 1]$ is the standard construction of the periodized multiresolution analysis (Daubechies, 1992, Section 9.3). Since $\dim V_J = 2^J$ and the $2^J - 1$ wavelets of level below

J are orthogonal to the constant, W_J is exactly the set of elements of V_J with zero integral.

Injectivity. Suppose two parameter vectors give the same regression function almost surely. The functions $\beta_\ell^J(u) = c_\ell + \sum_m g_{\ell m}^J(u_m)$, with $g_{\ell m}^J = \sum_{jk} \theta_{\ell m, jk} \psi_{jk}$, are measurable and bounded, so the first part of Proposition 1 gives $\beta_\ell^J = \tilde{\beta}_\ell^J$ almost surely; each $g_{\ell m}^J$ has zero integral by the previous paragraph, so the second part gives $c_\ell = \tilde{c}_\ell$ and $\sum_{jk} (\theta - \tilde{\theta})_{\ell m, jk} \psi_{jk} = 0$ almost everywhere, whence $\theta = \tilde{\theta}$ by orthonormality. For the Gram matrix, $E \|\mathbf{Z}\|_2^2 < \infty$ because $E \|\mathbf{X}\|_2^2 < \infty$ and each wavelet is bounded, and for $a \neq 0$, $a^\top \Sigma a = E\{(a^\top \mathbf{Z})^2\} = \|f_a\|_{L_2(P)}^2 > 0$ by injectivity, where f_a is the regression function with parameter a .

Nullity when the scaling column is kept. With $\phi_{00} \equiv 1$ the column $X_\ell \phi_{00}(U_m)$ coincides with the unpenalized column X_ℓ , for each of the pq pairs, so the pq vectors that oppose one to the other lie in the kernel and are linearly independent. There are no others: a kernel vector defines $\sum_\ell X_\ell \{a_\ell + \sum_m (\alpha_{\ell m} + \sum_{jk} a_{\ell m, jk} \psi_{jk}(U_m))\} = 0$ almost surely; by Proposition 1 applied to the part with zero integral, $a_{\ell m, jk} = 0$ and $a_\ell + \sum_m \alpha_{\ell m} = 0$, which is the space of dimension pq spanned by those vectors. $\square$

Remark S2.2 (The interval basis). With the basis of Cohen, Daubechies, and Vial (1993) the wavelets of level $j \geq j_0$ are orthogonal to V_{j_0} , which con-

tains the constants, so they still have zero integral and the penalized part of each block satisfies (2.2) with no restriction. The 2^{j_0} scaling functions do not: the constant is the combination $1 = \sum_k \mu_k \phi_{j_0 k}^{\text{int}}$ with $\mu_k = \int_0^1 \phi_{j_0 k}^{\text{int}}$, and the subspace with zero integral is $\{\mu^\top \alpha = 0\}$. Keeping the 2^{j_0} scaling columns of each block unrestricted repeats the constant and makes Σ singular with nullity pq , as in the last part of Lemma 1; reparametrizing $\alpha_{\ell m} = Q \gamma_{\ell m}$ with Q an orthonormal basis of $\{\mu^\top \alpha = 0\}$ restores injectivity, at the price of $pq(2^{j_0} - 1)$ unpenalized parameters.

S3 Approximation by the wavelet spaces W_J

Proof of Lemma 2

Fix a pair (ℓ, m) and write $g = g_{\ell m}$ and $\theta_{jk} = \theta_{\ell m, jk}^*$.

Tail sum. By Step 2 below the coefficients are square summable, so $g \in L_2[0, 1]$ and, the system being complete and g orthogonal to the constant, $g = \sum_{j \geq 0} \sum_k \theta_{jk} \psi_{jk}$ in L_2 . The orthogonal projection onto W_J retains the levels below J , and Parseval's identity gives $\|g - \Pi_J g\|_{L_2[0,1]}^2 = \sum_{j \geq J} \|\theta_{j \cdot}\|_2^2$.

From ℓ_π to ℓ_2 inside a level. In $\mathbb{R}^{2^j}$ one has $\|a\|_2 \leq \|a\|_\pi$ when $\pi \leq 2$, and $\|a\|_2 \leq (2^j)^{1/2-1/\pi} \|a\|_\pi$ by Hölder's inequality when $\pi > 2$. In both

cases $\|\theta_{j\cdot}\|_2 \leq 2^{j(1/2-1/\pi)_+} \|\theta_{j\cdot}\|_\pi$, and Assumption 5 gives

$$\|\theta_{j\cdot}\|_2 \leq C_g 2^{-j\{s+1/2-1/\pi-(1/2-1/\pi)_+\}} = C_g 2^{-js'},$$

since the exponent equals s when $\pi \geq 2$ and $s - (1/\pi - 1/2)$ when $\pi < 2$.

This is where the loss $(1/\pi - 1/2)_+$ appears.

Geometric sum and density. $\sum_{j \geq J} C_g^2 2^{-2js'} = C_g^2 2^{-2Js'} / (1 - 2^{-2s'})$, finite because $s' > 0$, which is the first display of Lemma 2. For the second, write $f - f_J = \sum_{\ell, m} X_\ell h_{\ell m}(U_m)$ with $h_{\ell m} = g_{\ell m} - \Pi_J g_{\ell m}$, the levels cancelling. By the Cauchy-Schwarz inequality over the pq terms and $|X_\ell| \leq B_X$,

$$\mathbb{E}\{(f - f_J)^2\} \leq pq \sum_{\ell, m} \mathbb{E}\{X_\ell^2 h_{\ell m}^2(U_m)\} \leq pq B_X^2 \sum_{\ell, m} \mathbb{E}\{h_{\ell m}^2(U_m)\},$$

and $\mathbb{E}\{h^2(U_m)\} = \int_0^1 h^2 p_{U_m} \leq C_U \|h\|_{L_2[0,1]}^2$, the marginal density being bounded by C_U because integrating the joint density over the other $q - 1$ coordinates of $[0, 1]^{q-1}$ does not increase the bound. One factor pq comes from the Cauchy-Schwarz inequality and the other from the sum. $\square$

The uniform version is used in Section S5 to bound the empirical norm of the bias without an appeal to Markov's inequality.

Lemma S3.1 (Uniform approximation error). *Under Assumptions 4 and 5 with $s > 1/\pi$, each component has a continuous version and, for every $J \geq 0$,*

$$\|g_{\ell m} - \Pi_J g_{\ell m}\|_\infty \leq \frac{(L-1) \|\psi\|_\infty C_g}{1 - 2^{-(s-1/\pi)}} 2^{-J(s-1/\pi)},$$

and therefore $\|\mathbf{b}\|_n \leq \|f - f_J\|_\infty \leq pqB_X(L-1) \|\psi\|_\infty C_g 2^{-J(s-1/\pi)} / \{1 - 2^{-(s-1/\pi)}\}$ for every sample.

Proof. At a point u and a level j at most $L-1$ periodized translates are nonzero, because $\psi(2^j u - k) \neq 0$ requires $2^j u - k \in (0, L-1)$, so $\sum_k |\psi_{jk}(u)| \leq (L-1)2^{j/2} \|\psi\|_\infty$. With $\max_k |\theta_{jk}| \leq \|\theta_j\|_\pi \leq C_g 2^{-j(s+1/2-1/\pi)}$ the level contributes at most $(L-1) \|\psi\|_\infty C_g 2^{-j(s-1/\pi)}$ in the supremum norm. Since $s > 1/\pi$ the series converges absolutely and uniformly, so its sum is continuous and agrees almost everywhere with g ; the tail from level J on is the geometric sum of the displayed bounds, and the statement for $f - f_J$ is the triangle inequality over the pq terms. $\square$

The next proposition is the one referred to in Section 2.2 of the paper: it quantifies what the periodized basis charges for a component that does not match at the two ends of the interval.

Proposition S3.1 (Cost of periodization). *Let the basis be as in Assumption 4 and let $g \in B_{\pi,r}^s[0,1]$, the Besov space of the interval and not of the circle, with $s > 1/\pi$, $N > s$ and $\int_0^1 g = 0$. Then there is a constant C depending on g , on ψ and on (s, π) such that, for every J with $2^J \geq L-1$,*

$$\|g - \Pi_J g\|_{L_2[0,1]}^2 \leq C 2^{-2Js'} + 2(L-1) \|\psi\|_1^2 \|g\|_\infty^2 2^{-J},$$

so that the L_2 rate is $2^{-J \min(s', 1/2)}$. If g is continuous with $g(0) \neq g(1)$, the

second term is of the exact order 2^{-J} and $\|g - \Pi_J g\|_\infty$ does not converge to zero.

Proof. Let $\tilde{g}$ be the one periodic extension of g , so that $\theta_{jk} = \int_{\mathbb{R}} \tilde{g}(y) 2^{j/2} \psi(2^j y - k) dy$. At a level with $2^j \geq L-1$, separate the interior translates, whose support lies inside $[0, 1]$, from the at most $L-1$ boundary translates that cross the identification of 0 with 1. For an interior translate, $\theta_{jk} = \langle Eg, \psi_{jk} \rangle_{\mathbb{R}}$ for any extension Eg of g to the line; taking E a bounded extension operator from $B_{\pi,r}^s[0, 1]$ to $B_{\pi,r}^s(\mathbb{R})$ (Triebel, 1983, Chapter 3) and using the wavelet characterization of the latter, which for the implication from the function to the sequence requires only $N > s$ vanishing moments, gives $\|(\theta_{jk})_{\text{int}}\|_\pi \leq C_1 2^{-j(s+1/2-1/\pi)}$, whose tail sum is bounded as in Lemma 2. For a boundary translate, $s > 1/\pi$ makes g bounded and $|\theta_{jk}| \leq \|g\|_\infty \|\psi_{jk}\|_1 = \|g\|_\infty \|\psi\|_1 2^{-j/2}$, and there are at most $L-1$ of them per level, so their tail sum is at most $2(L-1) \|\psi\|_1^2 \|g\|_\infty^2 2^{-J}$. For the exact order, if g is continuous with $g(0) \neq g(1)$ then on the support of a boundary translate $\tilde{g}$ equals $g(1) + o(1)$ to the left of the identification point and $g(0) + o(1)$ to the right, so $\theta_{jk} = 2^{-j/2} \{g(0) - g(1)\} \int_{y>a_k} \psi(y) dy + o(2^{-j/2})$; since $\int \psi = 0$ and ψ vanishes on no subinterval of its support, that partial integral is nonzero for some translate, and the boundary contribution of the level is of order 2^{-j} . The failure of uniform convergence is that of any

wavelet series at a jump. $\square$

Under Assumption 5 nothing changes, since that assumption is already a periodic one; what Proposition S3.1 gives is the price of its failure. With the basis of Cohen, Daubechies, and Vial (1993) the space V_J reproduces polynomials of degree below N and its basis characterizes $B_{\pi,r}^s[0, 1]$ without periodicity (Cohen, 2003, Section 3.9), so Lemma 2 holds with that space and the same proof.

S4 The Gram matrix of the product design

Recall $\Psi(u) \in \mathbb{R}^K$, $K = 1 + qN_J$, the vector that collects the constant and the wavelets of the q modulators, and the identity $\mathbf{Z} = \Pi\{\mathbf{X} \otimes \Psi(\mathbf{U})\}$, where Π is the fixed permutation that moves the p entries $X_\ell \cdot 1$ to the front. Since Π is orthogonal, every statement about eigenvalues is invariant to the column order. Assumption 4 also gives a constant C_ψ , depending only on the family, with

$$\sup_{u \in [0,1]} \sum_{j < J} \sum_k \psi_{jk}(u)^2 \leq C_\psi 2^J \quad \text{for every } J \geq 1 : \quad (\text{S4.1})$$

at each level at most L periodized wavelets are nonzero at a given u and $\|\psi_{jk}\|_\infty = 2^{j/2} \|\psi\|_\infty$, so the left side is at most $L \|\psi\|_\infty^2 \sum_{j < J} 2^j \leq L \|\psi\|_\infty^2 2^J$.

Proof of Proposition 2

Part (i), the modulators. For $a \in \mathbb{R}^K$ write $h_a(u) = a^\top \Psi(u) = a_0 + \sum_m h_{a,m}(u_m)$ with $h_{a,m} = \sum_{jk} a_{m,jk} \psi_{jk}$. The functions 1 and $\psi_{jk}(u_m)$, $m = 1, \dots, q$, are orthonormal in $L_2([0, 1]^q, du)$: within one coordinate this is the orthonormality of the periodized system, and across coordinates $\int \psi_{jk}(u_m) \psi_{j'k'}(u_{m'}) du = \int_0^1 \psi_{jk} \int_0^1 \psi_{j'k'} = 0$ for $m \neq m'$, because every wavelet integrates to zero. Hence $\int_{[0,1]^q} h_a^2 = \|a\|_2^2$ and, by Assumption 3, $c_U \|a\|_2^2 \leq \mathbb{E}\{h_a(\mathbf{U})^2\} \leq C_U \|a\|_2^2$, which is the claim since $a^\top \Sigma_\Psi a = \mathbb{E}\{h_a(\mathbf{U})^2\}$.

Part (i), the full matrix. Let $\delta \in \mathbb{R}^{pK}$ and reshape it as $\Delta \in \mathbb{R}^{p \times K}$, row ℓ holding the coefficients attached to X_ℓ , so that $\{\mathbf{X} \otimes \Psi(\mathbf{U})\}^\top \delta = \mathbf{X}^\top \Delta \Psi(\mathbf{U})$. By the tower property,

$$\delta^\top \Pi^\top \Sigma \Pi \delta = \mathbb{E}[\{\mathbf{X}^\top \Delta \Psi(\mathbf{U})\}^2] = \mathbb{E}[\Psi(\mathbf{U})^\top \Delta^\top \mathbb{E}(\mathbf{X} \mathbf{X}^\top \mid \mathbf{U}) \Delta \Psi(\mathbf{U})].$$

Assumption 2 bounds the inner quadratic form between $\kappa_1 \|\Delta \Psi(\mathbf{U})\|_2^2$ and $\kappa_2 \|\Delta \Psi(\mathbf{U})\|_2^2$ almost surely, and $\mathbb{E} \|\Delta \Psi(\mathbf{U})\|_2^2 = \delta^\top (I_p \otimes \Sigma_\Psi) \delta$ lies between $\lambda_{\min}(\Sigma_\Psi) \|\delta\|_2^2$ and $\lambda_{\max}(\Sigma_\Psi) \|\delta\|_2^2$. Combining the two and using the first part gives $\gamma = \kappa_1 c_U \leq \lambda_{\min}(\Sigma)$ and $\lambda_{\max}(\Sigma) \leq \kappa_2 C_U$. If $\mathbf{X}$ and $\mathbf{U}$ are independent, the conditional second moment matrix is constant, the same computation gives $\delta^\top \Pi^\top \Sigma \Pi \delta = \delta^\top \{\mathbb{E}(\mathbf{X} \mathbf{X}^\top) \otimes \Sigma_\Psi\} \delta$ for every δ , and two

symmetric matrices with the same quadratic form are equal; the eigenvalues of a Kronecker product of symmetric matrices are the products of the eigenvalues, and $\lambda_{\min}\{\mathbb{E}(\mathbf{X}\mathbf{X}^\top)\} \geq \kappa_1$ by Jensen's inequality, since $\lambda_{\min}$ is concave and $\mathbb{E}(\mathbf{X}\mathbf{X}^\top) = \mathbb{E}\{\mathbb{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\}$.

Part (ii), the empirical matrix. First the range: $\|\mathbf{Z}_i\|_2^2 = \|\mathbf{X}_i\|_2^2 \|\boldsymbol{\Psi}(\mathbf{U}_i)\|_2^2 \leq pB_X^2(1 + qC_\psi 2^J) = R_J$ by (S4.1) applied to each of the q coordinates. Set $A_i = (\mathbf{Z}_i\mathbf{Z}_i^\top - \Sigma)/n$, independent, centered and symmetric, with $\widehat{\Sigma} - \Sigma = \sum_i A_i$. Then $\|\mathbf{Z}_i\mathbf{Z}_i^\top\|_{\text{op}} = \|\mathbf{Z}_i\|_2^2 \leq R_J$ and $\|\Sigma\|_{\text{op}} \leq R_J$, so $\|A_i\|_{\text{op}} \leq 2R_J/n$; and $\mathbb{E}\{(\mathbf{Z}\mathbf{Z}^\top - \Sigma)^2\} = \mathbb{E}(\|\mathbf{Z}\|_2^2 \mathbf{Z}\mathbf{Z}^\top) - \Sigma^2 \preceq R_J\Sigma$, so that $\|\sum_i \mathbb{E}A_i^2\|_{\text{op}} \leq R_J\lambda_{\max}(\Sigma)/n \leq R_J\kappa_2 C_U/n$. The matrix Bernstein inequality (Tropp, 2012, Theorem 1.4) applied to $\pm \sum_i A_i$ gives, for every $t > 0$,

$$\mathbb{P}(\|\widehat{\Sigma} - \Sigma\|_{\text{op}} \geq t) \leq 2D \exp\left\{-\frac{nt^2}{2R_J(\kappa_2 C_U + 2t/3)}\right\},$$

and $t = \gamma/2$ with Weyl's inequality gives $\lambda_{\min}(\widehat{\Sigma}) \geq \lambda_{\min}(\Sigma) - \gamma/2 \geq \gamma/2$ on the complementary event, which is the display of part (ii). The exponent is $-cn/R_J$ with c fixed and $R_J \asymp pq2^J$, while $\log(2D) \asymp \log(pq2^J)$, so the bound tends to zero exactly when $n/(pq2^J) - \log(pq2^J) \rightarrow \infty$, which is implied by $pq2^J \log(pq2^J)/n \rightarrow 0$. On that event $\phi_0^2(S; \widehat{\Sigma}) \geq \gamma/2$ for every support, by (S1.1). $\square$

Remark S4.1 (What the two assumptions exclude). Assumption 2 fails exactly when a nontrivial linear combination of the linear covariates is al-

most surely a function of the modulators on a set of positive probability; in the canonical case $X_1 \equiv 1$ and $X_2 = h(U_1)$ the conditional second moment matrix has rank one and the population Gram matrix degenerates as J grows, because $h(u_1)\psi_{jk}(u_1)$ is approached by the span of the basis. In Assumption 3 it is the joint density that is bounded below: if $U_2 = U_1$ almost surely, both marginals are uniform but $\psi_{jk}(U_1) - \psi_{jk}(U_2) \equiv 0$ and Σ_{Ψ} is singular. Independence between $\mathbf{X}$ and $\mathbf{U}$ buys the exact factorization of part (i) and nothing else.

S5 The oracle inequality

[k = 4: the calibration lemma of the lasso, the proof of Theorem 2 for the lasso, the remark on the cone and the corollary that instantiates the design condition, which stood in this section, are moved to Section S8, unchanged except for the numbering of the results they cite; Lemmas S5.1, S5.3 and S5.4 stay, the first extended to the penalty of equation (2.5).]

~~Four lemmas precede the proof of Theorem 2.~~ A proposition on the partition into chunks and four lemmas precede the proof of Theorem 2.

Proposition S5.1 (The partition into chunks). *Let $\mathcal{G}$ be the partition of Section 2.3 of the paper with $b \geq 2$ and $J \geq 1$, and let $j^* = \max\{j \geq 0 : 2^j < b\}$.*

(i) Every chunk of a fine level, $j > j^*$, is contained in that level and has between b and $2b - 1$ coefficients. If $J > j^* + 1$, the coarse chunk has $2^{j^*+1} - 1$ coefficients, and $b - 1 \leq 2^{j^*+1} - 1 \leq 2b - 3$; if $J \leq j^* + 1$, each block is a single chunk of $2^J - 1$ coefficients.

(ii) $|\mathcal{G}| \leq pq(1 + 2^J/b)$.

(iii) With $w_G = |G|^{1/2}$, $\varrho^2 = \max_G |G| / \min_G |G| \leq (2b - 1)/(b - 1) \leq 3$.

Proof. All blocks have the same structure, so it suffices to look at one. The coarse chunk has $\sum_{j=0}^{\min(j^*, J-1)} 2^j = 2^{\min(j^*, J-1)+1} - 1$ coefficients. If $J \leq j^* + 1$, this is $2^J - 1$ and there is no fine level. Otherwise $2^{j^*} < b \leq 2^{j^*+1}$ and, b being an integer, $2^{j^*} \leq b - 1$, so that $b - 1 \leq 2^{j^*+1} - 1 \leq 2b - 3$. A fine level j has $k_j = \lfloor 2^j/b \rfloor \geq 1$ chunks, $k_j - 1$ of them with b coefficients and the last with $b + r_j$, where $r_j = 2^j - k_j b \in \{0, \dots, b - 1\}$; all lie between b and $2b - 1$, which is (i). For (ii), a block has $1 + \sum_{j=j^*+1}^{J-1} \lfloor 2^j/b \rfloor \leq 1 + (2^J - 2^{j^*+1})/b \leq 1 + 2^J/b$ chunks. For (iii), with $w_G = |G|^{1/2}$, $\varrho^2 = \max |G| / \min |G|$; with no fine level every chunk has $2^J - 1$ coefficients and $\varrho = 1$, and otherwise (i) gives $\min |G| \geq b - 1$ and $\max |G| \leq 2b - 1$, and $(2b - 1)/(b - 1) = 2 + 1/(b - 1) \leq 3$. $\square$

Lemma S5.1 (Profiling and the Schur complement). *Suppose $\mathbf{A}$ has rank p . Then $\hat{\boldsymbol{\theta}}$ minimizes $\left\| M_{\mathbf{A}} \mathbf{Y} - \tilde{\mathbf{B}} \boldsymbol{\theta} \right\|_n^2 + 2\lambda \|\boldsymbol{\theta}\|_1 \left\| M_{\mathbf{A}} \mathbf{Y} - \tilde{\mathbf{B}} \boldsymbol{\theta} \right\|_n^2 + 2\lambda \|\boldsymbol{\theta}\|_{\mathcal{G}, w}$,*

or the same with $\|\boldsymbol{\theta}\|_1$ for the lasso, and $\widehat{\mathbf{c}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top (\mathbf{Y} - \mathbf{B}\widehat{\boldsymbol{\theta}})$; with $v = \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$,

$$\widehat{f} - f_J = \widetilde{\mathbf{B}}v + \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}), \quad (\text{S5.1})$$

and the same identity holds with any comparator $\bar{\boldsymbol{\beta}}$, $\bar{\mathbf{b}}$ and $\bar{v} = \widehat{\boldsymbol{\theta}} - \bar{\boldsymbol{\theta}}$ in place of $\boldsymbol{\beta}^*$, $\mathbf{b}$ and v ; $\widehat{\Sigma}_{\mathbf{B}|\mathbf{A}}$ is the Schur complement of the block $\mathbf{A}$ in $\widehat{\Sigma}$ and

$$\widetilde{\gamma} = \lambda_{\min}(\widehat{\Sigma}_{\mathbf{B}|\mathbf{A}}) \geq \lambda_{\min}(\widehat{\Sigma}); \quad (\text{S5.2})$$

and if $\lambda_{\min}(\widehat{\Sigma}) > 0$ the minimizer in (2.5) is unique.

Proof. The argument uses only that the penalty does not depend on $\mathbf{c}$ and is convex, so it holds for both penalties. The objective is convex and, for fixed $\boldsymbol{\theta}$, quadratic and coercive in $\mathbf{c}$, with minimum at the stated value and minimized value $\|M_{\mathbf{A}}(\mathbf{Y} - \mathbf{B}\boldsymbol{\theta})\|_n^2 = \left\|M_{\mathbf{A}}\mathbf{Y} - \widetilde{\mathbf{B}}\boldsymbol{\theta}\right\|_n^2$, since $M_{\mathbf{A}}$ is a symmetric idempotent; minimizing first in $\mathbf{c}$ and then in $\boldsymbol{\theta}$ gives the same set of minimizers as minimizing jointly. For (S5.1), $\mathbf{A}\widehat{\mathbf{c}} = \Pi_{\mathbf{A}}(\mathbf{Y} - \mathbf{B}\widehat{\boldsymbol{\theta}})$ and $\mathbf{A}\mathbf{c} = \Pi_{\mathbf{A}}\mathbf{A}\mathbf{c}$, and $\mathbf{Y} = \mathbf{A}\mathbf{c} + \mathbf{B}\boldsymbol{\theta}^* + \mathbf{b} + \boldsymbol{\varepsilon}$, so $\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c}) = \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}) - \Pi_{\mathbf{A}}\mathbf{B}v$ and $\widehat{f} - f_J = M_{\mathbf{A}}\mathbf{B}v + \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon})$; the same computation with $\mathbf{Y} = \mathbf{A}\bar{\mathbf{c}} + \mathbf{B}\bar{\boldsymbol{\theta}} + \bar{\mathbf{b}} + \boldsymbol{\varepsilon}$ gives the identity for a comparator. For (S5.2), the variational characterization of the Schur complement gives, for every $v \in \mathbb{R}^d$,

$$v^\top \widehat{\Sigma}_{\mathbf{B}|\mathbf{A}} v = \min_{u \in \mathbb{R}^p} (u^\top, v^\top) \widehat{\Sigma} (u^\top, v^\top)^\top \geq \min_u \lambda_{\min}(\widehat{\Sigma}) (\|u\|_2^2 + \|v\|_2^2) = \lambda_{\min}(\widehat{\Sigma}) \|v\|_2^2.$$

Finally $\lambda_{\min}(\widehat{\Sigma}) > 0$ makes the quadratic term strictly convex, and adding a convex function preserves strict convexity. $\square$

Lemma S5.2 (Calibration of the penalty over chunks). *Under Assumption 1, let $\mathcal{G}$ be any partition of the penalized coordinates and $\alpha \in (0, 1)$.*

(i) *With $\lambda_{0,G}$ as in Theorem 2, the event $\mathcal{E} = \{\|(\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n)_G\|_2 \leq \lambda_{0,G} \text{ for every } G\}$ has $\mathbb{P}(\mathcal{E} \mid \mathbf{X}, \mathbf{U}) \geq 1 - \alpha$. Consequently, for any weights w and every $\lambda \geq \lambda_w$, $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$ and $\mathbb{P}(\mathcal{T}_{\mathcal{G},w}) \geq 1 - \alpha$.*

(ii) *(Closed form.) $\text{tr}(\widetilde{\Psi}_G) \leq |G|\widehat{\sigma}_{\max}^2$ and $\|\widetilde{\Psi}_G\|_{\text{op}} \leq \lambda_{\max}(\widehat{\Sigma})$; hence, on the event $\{\widehat{\sigma}_{\max}^2 \leq 2B_X^2 C_U\} \cap \{\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda\}$, $\lambda_{0,G} \leq \bar{\lambda}_{0,G}$ for every G , with*

$$\bar{\lambda}_{0,G} = \frac{\sigma}{\sqrt{n}} \left\{ B_X \sqrt{2C_U |G|} + \sqrt{2\Lambda \log(|\mathcal{G}|/\alpha)} \right\},$$

which is deterministic and is the quantity of equation (3.1) of the paper.

(iii) *(The price of the weights.) With $\lambda_n^{\mathcal{G}} = 2 \max_G \bar{\lambda}_{0,G}/w_G$ and $\lambda_{n,1}^{\mathcal{G}} = 2 \max_G \bar{\lambda}_{0,G}$, $\lambda_n^{\mathcal{G}} w_G \leq \varrho \lambda_{n,1}^{\mathcal{G}}$ for every G , and therefore $(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{H}) \leq \varrho^2 (\lambda_{n,1}^{\mathcal{G}})^2 |\mathcal{H}|$ for every $\mathcal{H} \subseteq \mathcal{G}$. The same holds with λ_w and $2 \max_G \lambda_{0,G}$ in place of $\lambda_n^{\mathcal{G}}$ and $\lambda_{n,1}^{\mathcal{G}}$.*

The event $\mathcal{E}$ does not depend on the weights: they only choose the chunk that dictates the penalty, the one with the largest $\lambda_{0,G}/w_G$, and the price of unequal weights is the factor ϱ of (iii), paid in the estimation

term. With unit weights and chunks of at most $b_{\max}$ coefficients, $\lambda_{n,1}^{\mathcal{G}} \asymp \sigma\{(b_{\max} + \log |\mathcal{G}|)/n\}^{1/2}$, which is the penalty level of Klopp and Pensky (2015, eq. 3.14) up to constants and the sub-Gaussian version of the level of Lounici et al. (2011, Lemma 3.1).

Proof. (i). Condition on $(\mathbf{X}, \mathbf{U})$. Then $\tilde{\mathbf{B}}$ is fixed and, by Assumption 1, the errors are independent with $\mathbb{E}\{\exp(a_i \varepsilon_i) \mid \mathbf{X}, \mathbf{U}\} \leq \exp(\sigma^2 a_i^2/2)$, so that $\mathbb{E}\{\exp(a^\top \boldsymbol{\varepsilon}) \mid \mathbf{X}, \mathbf{U}\} \leq \exp(\sigma^2 \|a\|_2^2/2)$ for every $a \in \mathbb{R}^n$, which is the condition of Hsu et al. (2012, Theorem 2.1) with mean zero. Fix G and write $M_G = \tilde{\mathbf{B}}_G^\top/n$, so that $(\tilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n)_G = M_G \boldsymbol{\varepsilon}$, and $\Sigma_G = M_G^\top M_G \in \mathbb{R}^{n \times n}$. The theorem is stated for a square matrix A with $\Sigma = A^\top A$; take $A = \Sigma_G^{1/2}$, which has the same Σ_G , and note that $\left\| \Sigma_G^{1/2} \boldsymbol{\varepsilon} \right\|_2^2 = \boldsymbol{\varepsilon}^\top \Sigma_G \boldsymbol{\varepsilon} = \|M_G \boldsymbol{\varepsilon}\|_2^2$. Since $\text{tr}(\Sigma_G) = \text{tr}(\tilde{\Psi}_G)/n$, $\|\Sigma_G\|_{\text{op}} = \left\| \tilde{\Psi}_G \right\|_{\text{op}}/n$ and $\text{tr}(\Sigma_G^2) \leq \|\Sigma_G\|_{\text{op}} \text{tr}(\Sigma_G)$, the theorem gives, for every $t > 0$, with conditional probability at least $1 - e^{-t}$,

$$\left\| (\tilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n)_G \right\|_2^2 \leq \sigma^2 [\text{tr}(\Sigma_G) + 2\{\text{tr}(\Sigma_G^2)t\}^{1/2} + 2\|\Sigma_G\|_{\text{op}}t] \leq \frac{\sigma^2}{n} \left[\{\text{tr}(\tilde{\Psi}_G)\}^{1/2} + \{2\left\| \tilde{\Psi}_G \right\|_{\text{op}}t\}^{1/2} \right]^2.$$

With $t = \log(|\mathcal{G}|/\alpha)$ each chunk fails with conditional probability at most $\alpha/|\mathcal{G}|$, and a union bound over the chunks and the tower property give $\mathbb{P}(\mathcal{E}) \geq 1 - \alpha$. On $\mathcal{E}$, $\left\| (\tilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n)_G \right\|_2/w_G \leq \lambda_{0,G}/w_G \leq \lambda_w/2 \leq \lambda/2$ for every G , so $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$.

(ii). Since $M_{\mathbf{A}} \preceq I_n$, $\tilde{\Psi}_G = \mathbf{B}_G^\top M_{\mathbf{A}} \mathbf{B}_G/n \preceq \hat{\Sigma}_{GG}$, the principal sub-

matrix of $\widehat{\Sigma}$ on G . Hence $\text{tr}(\widetilde{\Psi}_G) \leq \sum_{a \in G} (\widehat{\Sigma})_{aa} \leq |G| \widehat{\sigma}_{\max}^2$, and $\|\widetilde{\Psi}_G\|_{\text{op}} \leq \lambda_{\max}(\widehat{\Sigma}_{GG}) \leq \lambda_{\max}(\widehat{\Sigma})$ by Cauchy interlacing. On the event of the statement, $\widehat{\sigma}_{\max} \leq B_X(2C_U)^{1/2}$ and $\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda$.

(iii). For every G , $\lambda_n^{\mathcal{G}} w_G = 2 \max_{G'} (\bar{\lambda}_{0,G'} w_G / w_{G'}) \leq 2 \max_{G'} \bar{\lambda}_{0,G'} \cdot \max_{G'} (w_G / w_{G'}) \leq \varrho \lambda_{n,1}^{\mathcal{G}}$, and $(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{H}) = \sum_{G \in \mathcal{H}} (\lambda_n^{\mathcal{G}} w_G)^2 \leq \varrho^2 (\lambda_{n,1}^{\mathcal{G}})^2 |\mathcal{H}|$.

The same computation holds with the $\lambda_{0,G}$. $\square$

Lemma S5.3 (Empirical column norms). *Under Assumptions 2 to 4, every penalized column has $\mathbb{E}\{(\widehat{\Sigma})_{aa}\} \leq B_X^2 C_U$ and, for every $t > 0$, with $R'_J = B_X^2 \|\psi\|_{\infty}^2 2^{J-1}$,*

$$\mathbb{P}(\widehat{\sigma}_{\max}^2 > B_X^2 C_U + t) \leq d \exp\left\{-\frac{nt^2}{2(R'_J B_X^2 C_U + R'_J t/3)}\right\}.$$

In particular $\mathbb{P}(\widehat{\sigma}_{\max}^2 \leq 2B_X^2 C_U) \rightarrow 1$ whenever $2^J \log d/n \rightarrow 0$.

Proof. For the column $a = (\ell m, jk)$ the summand is $W_i = X_{i\ell}^2 \psi_{jk}(U_{im})^2 \geq 0$, with $\mathbb{E}(W) \leq B_X^2 \int_0^1 \psi_{jk}^2 p_{U_m} \leq B_X^2 C_U$ by orthonormality and the bounded marginal density, and $0 \leq W_i \leq B_X^2 \|\psi\|_{\infty}^2 2^{J-1} = R'_J$ because $j \leq J-1$. Then $\text{Var}(W) \leq \mathbb{E}(W^2) \leq R'_J \mathbb{E}(W) \leq R'_J B_X^2 C_U$, and the one sided Bernstein inequality with a union bound over the d columns gives the display. With $t = B_X^2 C_U$ the exponent is $-3nB_X^2 C_U / (8R'_J)$ and $R'_J \asymp 2^J$. $\square$

Lemma S5.4 (The ℓ_1 norm of the [oracle](#)). *Under Assumption 5, for every pair (ℓ, m) and every level, $\sum_k |\theta_{\ell m, jk}^*| \leq 2^{j(1-1/\pi)} \|\theta_{\ell m, j\cdot}^*\|_{\pi} \leq C_g 2^{j(1/2-s)}$,*

with no dependence on π , and therefore $\|\boldsymbol{\theta}^*\|_1 \leq pqC_g \sum_{j < J} 2^{j(1/2-s)}$, which is $O(1)$ if $s > 1/2$, of order J if $s = 1/2$ and of order $2^{J(1/2-s)}$ if $s < 1/2$.

Proof. Hölder's inequality in $\mathbb{R}^{2^j}$ gives $\|a\|_1 \leq (2^j)^{1-1/\pi} \|a\|_\pi$, and Assumption 5 bounds the ℓ_π norm of the level by $C_g 2^{-j(s+1/2-1/\pi)}$; the resulting exponent is $j(1-1/\pi) - j(s+1/2-1/\pi) = j(1/2-s)$, in which π has disappeared. Summing over the levels below J and over the pq blocks gives the three regimes of the geometric sum. $\square$

Proof of Theorem 2

The event $\mathcal{T}_{\mathcal{G},w}$ has probability at least $1-\alpha$ whenever $\lambda \geq \lambda_w$, by Lemma S5.2(i).

Fix a comparator $\bar{\boldsymbol{\beta}} = (\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}})$ and write $\|\cdot\|$ for $\|\cdot\|_{\mathcal{G},w}$ in this proof.

Step 1, the basic inequality. By Lemma S5.1, $\hat{\boldsymbol{\theta}}$ minimizes the residualized objective; comparing its value at $\hat{\boldsymbol{\theta}}$ with its value at $\bar{\boldsymbol{\theta}}$, and using $\mathbf{Y} = \mathbf{A}\bar{\mathbf{c}} + \mathbf{B}\bar{\boldsymbol{\theta}} + \bar{\mathbf{b}} + \boldsymbol{\varepsilon}$ and $M_{\mathbf{A}}\mathbf{A} = 0$, so that $M_{\mathbf{A}}\mathbf{Y} = \tilde{\mathbf{B}}\bar{\boldsymbol{\theta}} + r$ with $r = M_{\mathbf{A}}(\bar{\mathbf{b}} + \boldsymbol{\varepsilon})$,

$$\left\| \tilde{\mathbf{B}}\bar{\mathbf{v}} \right\|_n^2 + 2\lambda \left\| \hat{\boldsymbol{\theta}} \right\| \leq 2\langle r, \tilde{\mathbf{B}}\bar{\mathbf{v}} \rangle_n + 2\lambda \left\| \bar{\boldsymbol{\theta}} \right\|. \quad (\text{S5.3})$$

Step 2, the two noises. Since $M_{\mathbf{A}}$ is symmetric and $M_{\mathbf{A}}\tilde{\mathbf{B}} = \tilde{\mathbf{B}}$, $\langle r, \tilde{\mathbf{B}}\bar{\mathbf{v}} \rangle_n = \langle \bar{\mathbf{b}} + \boldsymbol{\varepsilon}, \tilde{\mathbf{B}}\bar{\mathbf{v}} \rangle_n$. The stochastic part is handled by the dual norm of $\|\cdot\|$, which

is $z \mapsto \max_G \|z_G\|_2 / w_G$: on $\mathcal{T}_{\mathcal{G},w}$,

$$2|\langle \varepsilon, \tilde{\mathbf{B}}\bar{v} \rangle_n| = 2\left|\sum_G \langle (\tilde{\mathbf{B}}^\top \varepsilon/n)_G, \bar{v}_G \rangle\right| \leq 2 \max_G \frac{\|(\tilde{\mathbf{B}}^\top \varepsilon/n)_G\|_2}{w_G} \sum_G w_G \|\bar{v}_G\|_2 \leq \lambda \|\bar{v}\|.$$

The deterministic part is handled by the Cauchy-Schwarz and Young inequalities: $2|\langle \bar{\mathbf{b}}, \tilde{\mathbf{B}}\bar{v} \rangle_n| = 2|\langle M_{\mathbf{A}}\bar{\mathbf{b}}, \tilde{\mathbf{B}}\bar{v} \rangle_n| \leq 2\|\bar{\mathbf{b}}\|_n \|\tilde{\mathbf{B}}\bar{v}\|_n \leq \|\tilde{\mathbf{B}}\bar{v}\|_n^2/2 + 2\|\bar{\mathbf{b}}\|_n^2$. Substituting both in (S5.3),

$$\frac{1}{2} \|\tilde{\mathbf{B}}\bar{v}\|_n^2 + 2\lambda \|\hat{\boldsymbol{\theta}}\| \leq \lambda \|\bar{v}\| + 2\lambda \|\bar{\boldsymbol{\theta}}\| + 2\|\bar{\mathbf{b}}\|_n^2. \quad (\text{S5.4})$$

Step 3, the slow rate. Bounding $\|\bar{v}\| \leq \|\hat{\boldsymbol{\theta}}\| + \|\bar{\boldsymbol{\theta}}\|$ in (S5.4) gives part (i). No property of $\hat{\Sigma}$ was used.

Step 4, decomposability. For $G \notin \bar{\mathcal{G}}$, $\bar{\boldsymbol{\theta}}_G = 0$ and $\hat{\boldsymbol{\theta}}_G = \bar{v}_G$; for $G \in \bar{\mathcal{G}}$, $\|\hat{\boldsymbol{\theta}}_G\|_2 \geq \|\bar{\boldsymbol{\theta}}_G\|_2 - \|\bar{v}_G\|_2$. Hence $\|\hat{\boldsymbol{\theta}}\| \geq \|\bar{\boldsymbol{\theta}}\| - \|\bar{v}_{\bar{\mathcal{G}}}\| + \|\bar{v}_{\bar{\mathcal{G}}^c}\|$, while $\|\bar{v}\| = \|\bar{v}_{\bar{\mathcal{G}}}\| + \|\bar{v}_{\bar{\mathcal{G}}^c}\|$; substituting in (S5.4) and cancelling $2\lambda \|\bar{\boldsymbol{\theta}}\|$,

$$\frac{1}{2} \|\tilde{\mathbf{B}}\bar{v}\|_n^2 + \lambda \|\bar{v}_{\bar{\mathcal{G}}^c}\| \leq 3\lambda \|\bar{v}_{\bar{\mathcal{G}}}\| + 2\|\bar{\mathbf{b}}\|_n^2. \quad (\text{S5.5})$$

Step 5, two cases, with no cone. By the Cauchy-Schwarz inequality and the definition of $\tilde{\gamma}$,

$$\|\bar{v}_{\bar{\mathcal{G}}}\| = \sum_{G \in \bar{\mathcal{G}}} w_G \|\bar{v}_G\|_2 \leq \bar{\mathcal{W}}^{1/2} \|\bar{v}\|_2 \leq (\bar{\mathcal{W}}/\tilde{\gamma})^{1/2} \|\tilde{\mathbf{B}}\bar{v}\|_n, \quad (\text{S5.6})$$

because $\|\tilde{\mathbf{B}}\bar{v}\|_n^2 = \bar{v}^\top \hat{\Sigma}_{\mathbf{B}|\mathbf{A}} \bar{v} \geq \tilde{\gamma} \|\bar{v}\|_2^2$. If $2\|\bar{\mathbf{b}}\|_n^2 \leq \lambda \|\bar{v}_{\bar{\mathcal{G}}}\|$, then (S5.5) gives

$$\frac{1}{2} \|\tilde{\mathbf{B}}\bar{v}\|_n^2 + \lambda \|\bar{v}_{\bar{\mathcal{G}}^c}\| \leq 4\lambda \|\bar{v}_{\bar{\mathcal{G}}}\|, \text{ and with (S5.6), } \frac{1}{2} \|\tilde{\mathbf{B}}\bar{v}\|_n^2 \leq 4\lambda (\bar{\mathcal{W}}/\tilde{\gamma})^{1/2} \|\tilde{\mathbf{B}}\bar{v}\|_n,$$

that is $\left\|\tilde{\mathbf{B}}\bar{v}\right\|_n^2 \leq 64\lambda^2\bar{\mathcal{W}}/\tilde{\gamma}$ and $\|\bar{v}\|_2^2 \leq \left\|\tilde{\mathbf{B}}\bar{v}\right\|_n^2/\tilde{\gamma} \leq 64\lambda^2\bar{\mathcal{W}}/\tilde{\gamma}^2$. If instead $2\left\|\bar{\mathbf{b}}\right\|_n^2 > \lambda\|\bar{v}_{\bar{\mathcal{G}}}\|$, then (S5.5) gives $\frac{1}{2}\left\|\tilde{\mathbf{B}}\bar{v}\right\|_n^2 < 8\left\|\bar{\mathbf{b}}\right\|_n^2$, that is $\left\|\tilde{\mathbf{B}}\bar{v}\right\|_n^2 < 16\left\|\bar{\mathbf{b}}\right\|_n^2$, and $\|\bar{v}\|_2^2 < 16\left\|\bar{\mathbf{b}}\right\|_n^2/\tilde{\gamma}$. Each quantity is bounded by the maximum of the two cases, hence by their sum, which is part (ii).

Step 6, prediction. By Lemma S5.1 with the comparator, $\hat{f} - f = \tilde{\mathbf{B}}\bar{v} + \Pi_{\mathbf{A}}(\bar{\mathbf{b}} + \boldsymbol{\epsilon}) - \bar{\mathbf{b}}$, and $\|\Pi_{\mathbf{A}}\bar{\mathbf{b}}\|_n \leq \left\|\bar{\mathbf{b}}\right\|_n$, which gives the triangle inequality of part (iii). Conditionally on the covariates the errors are independent, centered and of variance at most σ^2 , so $E(\|\Pi_{\mathbf{A}}\boldsymbol{\epsilon}\|_n^2 \mid \mathbf{X}, \mathbf{U}) \leq \sigma^2 \text{tr}(\Pi_{\mathbf{A}})/n \leq \sigma^2 p/n$. Since $\mathcal{T}_{\mathcal{G},w}$, $\tilde{\gamma}$ and λ do not depend on the comparator, the bounds hold simultaneously for all comparators, and with $(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$ and part (ii) they give, on $\mathcal{T}_{\mathcal{G},w} \cap \{\tilde{\gamma} > 0\}$,

$$\left\|\hat{f} - f\right\|_n^2 \leq 3 \min_{\bar{\beta}} \left\{ \frac{64\lambda^2\bar{\mathcal{W}}}{\tilde{\gamma}} + 20\left\|\bar{\mathbf{b}}\right\|_n^2 \right\} + 3\|\Pi_{\mathbf{A}}\boldsymbol{\epsilon}\|_n^2. \quad \square \quad (\text{S5.7})$$

Remark S5.1 (Why there is no cone condition, also over chunks). The step that dispenses with the cone is the one of the lasso (Remark S8.1): profiling removes the levels from the problem, and the full smallest eigenvalue of Proposition 2 controls $\|\bar{v}_{\bar{\mathcal{G}}}\| \leq (\bar{\mathcal{W}}/\tilde{\gamma})^{1/2} \left\|\tilde{\mathbf{B}}\bar{v}\right\|_n$ over the whole space, by (S5.6). Lounici et al. (2011) need a restricted eigenvalue condition for groups on a weighted cone (their Assumption 3.1); here it is dominated by the full smallest eigenvalue, and the constant is the same for every $\bar{\mathcal{G}}$.

Remark S5.2 (The estimator that the software computes). The software standardizes the columns, orthonormalizes each chunk and penalizes the orthonormalized coordinates with the weights $|G|^{1/2}$. On the original scale its penalty is $\sum_G w_G \left\| Q_G^{1/2} \boldsymbol{\theta}_G \right\|_2$, with $Q_G = \mathbf{B}_G^\top M_1 \mathbf{B}_G / n$ the centered Gram matrix of the chunk, $M_1 = I_n - n^{-1} \mathbf{1} \mathbf{1}^\top$. Theorem 2 holds with any Q_G positive definite and measurable with respect to the covariates, with three changes: the event becomes $\{\max_G \left\| Q_G^{-1/2} (\tilde{\mathbf{B}}^\top \boldsymbol{\varepsilon} / n)_G \right\|_2 / w_G \leq \lambda/2\}$, by the dual norm; $\bar{\mathcal{W}}$ becomes $q_{\max} \bar{\mathcal{W}}$ in parts (ii) and (iii), with $q_{\max} = \max_G \lambda_{\max}(Q_G)$, because Step 5 reads $\sum_{G \in \bar{\mathcal{G}}} w_G \left\| Q_G^{1/2} \bar{v}_G \right\|_2 \leq (q_{\max} \bar{\mathcal{W}})^{1/2} \|\bar{v}\|_2$; and Lemma S5.2(i) holds with $Q_G^{-1/2} \tilde{\Psi}_G Q_G^{-1/2}$ in place of $\tilde{\Psi}_G$, by the same proof with $Q_G^{-1/2} \tilde{\mathbf{B}}_G^\top / n$ in place of $\tilde{\mathbf{B}}_G^\top / n$. When the constant covariate $X_1 \equiv 1$ is among the columns of $\mathbf{A}$, $\Pi_{\mathbf{A}} \succeq I_n - M_1$, so that $M_{\mathbf{A}} \preceq M_1$ and $\tilde{\Psi}_G \preceq Q_G$: the trace of $Q_G^{-1/2} \tilde{\Psi}_G Q_G^{-1/2}$ is at most $|G|$ and its norm at most one, and

$$\lambda_{0,G} \leq \frac{\sigma}{\sqrt{n}} \left\{ |G|^{1/2} + \sqrt{2 \log(|\mathcal{G}|/\alpha)} \right\},$$

which is pivotal, with neither $\hat{\sigma}_{\max}$ nor Λ ; and $q_{\max} \leq \lambda_{\max}(\hat{\Sigma}) \leq \Lambda$ on $\mathcal{E}_\Sigma$, since $Q_G \preceq \hat{\Sigma}_{GG}$. The rates of Section S6 hold for this estimator with the constant multiplied by Λ .

Corollary S5.1 (The ideal risk form). *In the conditions of Theorem 2, on*

the event $\mathcal{T}_{\mathcal{G},w} \cap \{\tilde{\gamma} > 0\}$, let $\Lambda' \geq \lambda_{\max}(\widehat{\Sigma})$, for instance $\Lambda' = \Lambda$ on $\mathcal{E}_{\Sigma}$, and

$$\eta = \frac{1.6 \max_G (\lambda w_G)^2}{\Lambda' \tilde{\gamma}}.$$

Then

$$\left\| \widehat{f} - f \right\|_n^2 \leq 120 \Lambda' R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta) + 120 \|\mathbf{b}\|_n^2 + 3 \|\Pi_{\mathbf{A}} \boldsymbol{\varepsilon}\|_n^2, \quad \left\| \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \right\|_2^2 \leq \frac{82 \Lambda' R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta) + 64 \|\mathbf{b}\|_n^2}{\tilde{\gamma}}.$$

With $\lambda = \lambda_n^{\mathcal{G}}$, or $\lambda = \lambda_w$, Lemma S5.2(iii) gives $\eta \leq \varrho^2 \eta_1$, with $\eta_1 = 1.6(\lambda_{n,1}^{\mathcal{G}})^2/(\Lambda' \tilde{\gamma})$ the level of unit weights, and $R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta) \leq \varrho^2 R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta_1)$.

Proof. Let $T = \{G : \|\boldsymbol{\theta}_G^*\|_2^2 > \eta\}$ and take the comparator $\bar{\boldsymbol{\beta}} = (\mathbf{c}, \bar{\boldsymbol{\theta}})$ with $\bar{\boldsymbol{\theta}}_G = \boldsymbol{\theta}_G^* \mathbf{1}\{G \in T\}$. Then $\bar{\mathcal{G}} \subseteq T$ and, by the definition of η , $\lambda^2 \bar{\mathcal{W}} \leq \sum_{G \in T} (\lambda w_G)^2 \leq |T| \max_G (\lambda w_G)^2 = |T| \eta \Lambda' \tilde{\gamma} / 1.6$. The residual is $\bar{\mathbf{b}} = \mathbf{b} + \mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})$, and since $\mathbf{B}^\top \mathbf{B} / n$ is a principal submatrix of $\widehat{\Sigma}$,

$$\|\bar{\mathbf{b}}\|_n^2 \leq 2 \|\mathbf{b}\|_n^2 + 2 \lambda_{\max}(\widehat{\Sigma}) \|\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}}\|_2^2 \leq 2 \|\mathbf{b}\|_n^2 + 2 \Lambda' \sum_{G \notin T} \|\boldsymbol{\theta}_G^*\|_2^2.$$

By (S5.7), $\left\| \widehat{f} - f \right\|_n^2 \leq 192 \lambda^2 \bar{\mathcal{W}} / \tilde{\gamma} + 60 \|\bar{\mathbf{b}}\|_n^2 + 3 \|\Pi_{\mathbf{A}} \boldsymbol{\varepsilon}\|_n^2 \leq 120 \Lambda' \{\eta |T| + \sum_{G \notin T} \|\boldsymbol{\theta}_G^*\|_2^2\} + 120 \|\mathbf{b}\|_n^2 + 3 \|\Pi_{\mathbf{A}} \boldsymbol{\varepsilon}\|_n^2$, and the term in braces is exactly $R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta)$. For the coefficients, Theorem 2(ii) gives $\|\bar{\mathbf{v}}\|_2^2 \leq 64 \lambda^2 \bar{\mathcal{W}} / \tilde{\gamma}^2 + 16 \|\bar{\mathbf{b}}\|_n^2 / \tilde{\gamma} \leq (40 \Lambda' \eta |T| + 32 \|\mathbf{b}\|_n^2 + 32 \Lambda' \sum_{G \notin T} \|\boldsymbol{\theta}_G^*\|_2^2) / \tilde{\gamma} \leq (40 \Lambda' R_{\mathcal{G}} + 32 \|\mathbf{b}\|_n^2) / \tilde{\gamma}$, and $\left\| \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \right\|_2^2 \leq 2 \|\bar{\mathbf{v}}\|_2^2 + 2 \|\bar{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|_2^2 \leq (80 \Lambda' R_{\mathcal{G}} + 64 \|\mathbf{b}\|_n^2) / \tilde{\gamma} + 2 R_{\mathcal{G}}$; since $\tilde{\gamma} \leq \lambda_{\min}(\mathbf{B}^\top \mathbf{B} / n) \leq \lambda_{\max}(\widehat{\Sigma}) \leq \Lambda'$, the Schur complement being dominated by the block, $2 R_{\mathcal{G}} \leq 2 \Lambda' R_{\mathcal{G}} / \tilde{\gamma}$, which gives the second bound.

The last statement is Lemma S5.2(iii) and the two properties of R_G in Section S1. $\square$

S6 Rates

[k = 4: the lemma that read Theorem 2 at an arbitrary comparator for the lasso, and the proofs of Lemma 3, Proposition 3, Theorem 3, Corollary 1, Theorem 1 and the remark on compressibility for the lasso, which stood in this section, are moved to Section S8, unchanged except for the numbering of the results they cite.]

Throughout this section $p, q, \sigma, B_X, c_U, C_U, \kappa_1, \kappa_2, C_g, s, \pi, r$ and α are fixed as n grows, $J = J_n \rightarrow \infty$, $d_n = pq(2^{J_n} - 1)$, $L_n = \log(2d_n/\alpha) \asymp J_n$, and λ_n is as in (3.1) $2^{J_n} \leq 2n$, $b_n = \lceil \log n \rceil$ with $n \geq 3$, so that $b_n \geq 2$, $\mathcal{G}_n$ is the partition of Section 2.3 of the paper with $b = b_n$, the weights have ratio $\varrho \leq \bar{\varrho}$ for a fixed $\bar{\varrho}$, and $\lambda = \lambda_n^{\mathcal{G}}$ is as in (3.1). Since the constants of Theorem 2 depend on the confidence levels only through $1/\alpha'$ and $1/\alpha''$, taking $\alpha' = \alpha'' = \epsilon/3$ gives, for each $\epsilon > 0$, a constant M_ϵ with $\mathbb{P}(\|\hat{f} - f\|_n^2 > M_\epsilon r_n) < \epsilon$, which is the meaning of the statements in O_p below. We also write $\Lambda = \kappa_2 C_U + \gamma/2$. The statements in O_p are proved by exhibiting, for each $\epsilon > 0$, an event of probability at least $1 - \epsilon$ for n large, built with the confidence levels $\alpha = \alpha' = \alpha'' = \epsilon/4$ for the noise, the bias

and the levels, on which the error is at most a constant M_ϵ times the rate.

The constant $\Lambda = \kappa_2 C_U + \gamma/2$ and the event $\mathcal{E}_\Sigma$ are those of Section S1.

Lemma S6.1 (The penalty level and the regime). *In the setting above,*

$$(\lambda_{n,1}^{\mathcal{G}})^2 \leq C_\lambda \frac{\sigma^2 b_n}{n}, \quad C_\lambda = 16\{2B_X^2 C_U + \Lambda(1 + \tfrac{1}{2} \log(2pq/\alpha))\}, \quad (\text{S6.1})$$

and, on the event $\mathcal{A}_n = \{\hat{\sigma}_{\max}^2 \leq 2B_X^2 C_U\} \cap \mathcal{E}_\Sigma$, $\lambda_n^{\mathcal{G}} \geq \lambda_w$, so that $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$

for the event $\mathcal{E}$ of Lemma S5.2(i), $\tilde{\gamma} > \gamma/2$, $\mathbf{A}$ has rank p and $\lambda_{\max}(\hat{\Sigma}) < \Lambda$.

Moreover $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$ whenever $pq2^{J_n} \log(pq2^{J_n})/n \rightarrow 0$.

Proof. By $(x+y)^2 \leq 2x^2+2y^2$, $(\lambda_{n,1}^{\mathcal{G}})^2 = 4 \max_G \bar{\lambda}_{0,G}^2 \leq (8\sigma^2/n)\{2B_X^2 C_U b_{\max} + 2\Lambda \log(|\mathcal{G}_n|/\alpha)\}$, with $b_{\max} = \max_G |G|$. By Proposition S5.1, $b_{\max} \leq 2b_n - 1 < 2b_n$ and $|\mathcal{G}_n| \leq pq(1 + 2^{J_n}/b_n) \leq pq(1 + n) \leq 2pqn$, because $2^{J_n} \leq 2n$ and $b_n \geq 2$; hence $\log(|\mathcal{G}_n|/\alpha) \leq \log(2pq/\alpha) + \log n \leq \log(2pq/\alpha) + b_n$, and $(\lambda_{n,1}^{\mathcal{G}})^2 \leq (16\sigma^2/n)\{(2B_X^2 C_U + \Lambda)b_n + \Lambda \log(2pq/\alpha)\} \leq C_\lambda \sigma^2 b_n/n$, using $b_n \geq 2$. On $\mathcal{A}_n$, Lemma S5.2(ii) gives $\lambda_{0,G} \leq \bar{\lambda}_{0,G}$ for every G , so that $\lambda_n^{\mathcal{G}} \geq \lambda_w$ and, by the same lemma, $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$. On $\mathcal{E}_\Sigma$, $\lambda_{\min}(\hat{\Sigma}) > \gamma/2$ and $\lambda_{\max}(\hat{\Sigma}) < \Lambda$; (S5.2) gives $\tilde{\gamma} > \gamma/2$, and $\lambda_{\min}(\mathbf{A}^\top \mathbf{A}/n) \geq \lambda_{\min}(\hat{\Sigma}) > 0$ gives the rank. Finally $\mathbb{P}(\hat{\sigma}_{\max}^2 \leq 2B_X^2 C_U) \rightarrow 1$ by Lemma S5.3, which asks $2^J \log d/n \rightarrow 0$, and $\mathbb{P}(\mathcal{E}_\Sigma) \rightarrow 1$ when $pq2^{J_n} \log(pq2^{J_n})/n \rightarrow 0$ (Section S1); the second condition implies the first. $\square$

Proof of Lemma 3

Fix a block and write $\theta_j = \theta_{\ell_{m,j}}^* \in \mathbb{R}^{2^j}$, set to zero for $j \geq J$, which only decreases the risk; write $\mathcal{G}_j$ for the chunks of a fine level j , those with $2^j \geq b$, and $r_j = \sum_{G \in \mathcal{G}_j} \min(\|\theta_G\|_2^2, \eta)$. The coarse chunk, if there is one, contributes at most η , so the block contributes at most $\eta + \sum_j r_j$, the sum running over the fine levels. The first bound on r_j is the count: each chunk of $\mathcal{G}_j$ has at least b of the 2^j coefficients of the level, so that $r_j \leq |\mathcal{G}_j| \eta \leq 2^j \eta / b$.

The case $\pi \leq 2$. Write $a = \pi(s + 1/2) > 1$. For $u \geq 0$, $\min(u, \eta) \leq \eta^{1-\pi/2} u^{\pi/2}$, and $\|x\|_2 \leq \|x\|_\pi$ for $\pi \leq 2$; hence the second bound,

$$r_j \leq \eta^{1-\pi/2} \sum_{G \in \mathcal{G}_j} \|\theta_G\|_\pi^\pi \leq \eta^{1-\pi/2} \|\theta_j\|_\pi^\pi \leq \eta^{1-\pi/2} C_g^\pi 2^{-j(a-1)},$$

by Assumption 5, because the chunks of $\mathcal{G}_j$ are disjoint and contained in the level. Define $x \in \mathbb{R}$ by $2^{xa} = b C_g^\pi \eta^{-\pi/2}$, so that the two bounds agree at $j = x$, with common value $h = 2^x \eta / b = \eta^{1-\pi/2} C_g^\pi 2^{-x(a-1)}$. Summing the first bound over the levels $j \leq x$ gives at most $2^{\lfloor x \rfloor + 1} \eta / b \leq 2h$ (an empty sum if $x < 0$), and summing the second over the levels $j > x$ gives at most $h \sum_{i \geq 0} 2^{-i(a-1)} = h / (1 - 2^{1-a})$, since the first integer $j > x$ has $2^{-j(a-1)} \leq 2^{-x(a-1)}$. The block therefore contributes at most $\eta + \{2 + (1 - 2^{1-a})^{-1}\} h$. Since $1/a = \tau/\pi$, $2^x = b^{\tau/\pi} C_g^\tau \eta^{-\tau/2}$ and $h = b^{\tau/\pi-1} C_g^\tau \eta^{1-\tau/2}$, and, with

$$1 - \tau/2 = 2s/(2s+1),$$

$$h = C_g^\tau \left(\frac{\eta}{b}\right)^{2s/(2s+1)} b^{\tau/\pi - 1 + 2s/(2s+1)} = C_g^\tau \left(\frac{\eta}{b}\right)^{2s/(2s+1)} b^{\tau(1/\pi - 1/2)}, \quad \tau\left(\frac{1}{\pi} - \frac{1}{2}\right) = \frac{2/\pi - 1}{2s+1}.$$

The case $\pi \geq 2$. Hölder's inequality in $\mathbb{R}^{2^j}$ gives $\|\theta_j\|_2 \leq 2^{j(1/2 - 1/\pi)} \|\theta_j\|_\pi \leq C_g 2^{-js}$, with $1/\pi = 0$ if $\pi = \infty$, and the second bound is $r_j \leq \|\theta_j\|_2^2 \leq C_g^2 2^{-2js}$. With $2^{x(2s+1)} = C_g^2 b/\eta$ and $h = 2^x \eta/b = C_g^2 2^{-2xs}$, the same computation gives the contribution $\eta + \{2 + (1 - 2^{-2s})^{-1}\}h$, and $h = (C_g^2 b/\eta)^{1/(2s+1)} \eta/b = C_g^\tau (\eta/b)^{2s/(2s+1)}$, because $2/(2s+1) = \tau$.

Summing over the pq blocks gives the statement; at $\pi = 2$, $1 - 2^{1-a} = 1 - 2^{-2s}$ and the two constants agree. $\square$

We also record that, on the event of Proposition 2(ii) with $t = \gamma/2$,

$$\|\mathbf{B}\delta\|_n^2 \leq \lambda_{\max}(\widehat{\Sigma}) \|\delta\|_2^2 \quad \text{and} \quad \lambda_{\max}(\widehat{\Sigma}) \leq \Lambda, \quad (\text{S6.2})$$

the first because $\mathbf{B}^\top \mathbf{B}/n$ is a principal submatrix of $\widehat{\Sigma}$ and the second by Weyl's inequality and Proposition 2(i).

Lemma S6.2 (The norm of the oracle over chunks). *Under Assumptions 4 and 5, let $\mathcal{G}$ be the partition of Section 2.3 of the paper with $b \geq 2$, and let $\vartheta_\pi = \min(1 - 1/\pi, 1/2)$, with $\vartheta_\infty = 1/2$. Then, for every J ,*

$$\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} \leq pq C_g \left\{ \frac{1}{1 - 2^{-s'}} + b^{-\vartheta_\pi} \sum_{j=j^*+1}^{J-1} 2^{j(1/2-s)} \right\}. \quad (\text{S6.3})$$

The bound for each fine level is attained, by a level whose energy is spread

evenly when $\pi \geq 2$, and by a level with one large coefficient in each chunk when $\pi \leq 2$.

Proof. Fix a block and write θ_j for its coefficients of level j . By Assumption 5, $\|\theta_j\|_2 \leq 2^{j(1/2-1/\pi)+} \|\theta_j\|_\pi \leq C_g 2^{-js'}$, by Hölder's inequality in $\mathbb{R}^{2^j}$ if $\pi \geq 2$ and by $\|\cdot\|_2 \leq \|\cdot\|_\pi$ if $\pi \leq 2$. The coarse chunk contributes at most $\sum_{j \leq j^*} \|\theta_j\|_2 \leq C_g(1-2^{-s'})^{-1}$. At a fine level j the $k_j \leq 2^j/b$ chunks $G \in \mathcal{G}_j$ partition the level. If $\pi \geq 2$, by the Cauchy-Schwarz inequality,

$$\sum_{G \in \mathcal{G}_j} \|\theta_G\|_2 \leq k_j^{1/2} \|\theta_j\|_2 \leq \left(\frac{2^j}{b}\right)^{1/2} C_g 2^{-js} = C_g b^{-1/2} 2^{j(1/2-s)},$$

if $\pi \leq 2$, by $\|\theta_G\|_2 \leq \|\theta_G\|_\pi$ and Hölder's inequality over the chunks,

$$\sum_{G \in \mathcal{G}_j} \|\theta_G\|_2 \leq k_j^{1-1/\pi} \left(\sum_{G \in \mathcal{G}_j} \|\theta_G\|_\pi^\pi \right)^{1/\pi} = k_j^{1-1/\pi} \|\theta_j\|_\pi \leq \left(\frac{2^j}{b}\right)^{1-1/\pi} C_g 2^{-j(s+1/2-1/\pi)} = C_g b^{-(1-1/\pi)} 2^{j(1/2-s)}.$$

Both are $C_g b^{-\vartheta_\pi} 2^{j(1/2-s)}$, and the equality cases of the two inequalities are the configurations of the statement. Summing over the fine levels $j^* < j < J$ and over the pq blocks gives (S6.3). $\square$

Proof of Proposition 3

The nonasymptotic bound. On $\mathcal{T}_{\mathcal{G},w}$, Theorem 2(i) with the comparator $\bar{\beta} = \beta^*$, so that $\bar{\mathbf{b}} = \mathbf{b}$ and $\bar{v} = v = \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$, gives $\left\| \tilde{\mathbf{B}}v \right\|_n^2 \leq 6\lambda \|\boldsymbol{\theta}^*\|_{\mathcal{G},w} + 4 \|\mathbf{b}\|_n^2$, and the first part of Theorem 2(iii), with $(x+y+z)^2 \leq 3(x^2+y^2+z^2)$,

gives

$$\left\| \widehat{f} - f \right\|_n^2 \leq 18\lambda \|\boldsymbol{\theta}^*\|_{\mathcal{G},w} + 24 \|\mathbf{b}\|_n^2 + 3 \|\Pi_{\mathbf{A}} \boldsymbol{\varepsilon}\|_n^2 \quad \text{on } \mathcal{T}_{\mathcal{G},w}. \quad (\text{S6.4})$$

Steps 1 to 3 and 6 of the proof of Theorem 2, the only ones used, do not involve $\widehat{\Sigma}$. With $\lambda \geq \lambda_w$, calibrated on the covariates by Lemma S5.2(i), the bound holds with probability at least $1 - \alpha$ in any design in which $\mathbf{A}$ has rank p , including designs with $d > n$, where $\widetilde{\gamma} = 0$.

The penalty level of the theory. Let $\mathcal{A}_n^+ = \{\widehat{\sigma}_{\max}^2 \leq 2B_X^2 C_U\} \cap \{\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda\}$. On it Lemma S5.2(ii) gives $\lambda_{0,G} \leq \bar{\lambda}_{0,G}$ for every G , hence $\lambda_n^{\mathcal{G}} \geq \lambda_w$ and $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$. The first event has probability tending to one by Lemma S5.3, whose proof uses only $|X_\ell| \leq B_X$ and $p_{\mathbf{U}} \leq C_U$. For the second, $\mathbb{P}\{\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda\} \geq \mathbb{P}\{\|\widehat{\Sigma} - \Sigma\|_{\text{op}} < \Lambda - \kappa_2 C_U\}$ by Weyl's inequality, since $\lambda_{\max}(\Sigma) \leq \kappa_2 C_U$, and the matrix Bernstein bound in the proof of Proposition 2(ii), with $t = \Lambda - \kappa_2 C_U$, sends it to one; that bound uses only $\|\mathbf{Z}_i\|_2^2 \leq R_J$ and $\lambda_{\max}(\Sigma) \leq \kappa_2 C_U$, the upper half of Proposition 2(i), and neither c_U nor κ_1 , so that any $\Lambda > \kappa_2 C_U$ serves. Both probabilities tend to one when $pq2^{J_n} \log(pq2^{J_n})/n \rightarrow 0$. Fix $\epsilon > 0$, take $\alpha = \alpha' = \alpha'' = \epsilon/4$ and intersect $\mathcal{A}_n^+$, $\mathcal{E}$, $\{\|\mathbf{b}\|_n^2 \leq \mathcal{B}_{J_n}^2/\alpha'\}$ and $\{\|\Pi_{\mathbf{A}} \boldsymbol{\varepsilon}\|_n^2 \leq \sigma^2 p/(n\alpha'')\}$, the last two by Markov's inequality, Lemma 2 and Theorem 2(iii); for n large the intersection has probability at least $1 - \epsilon$, and on it $\left\| \widehat{f} - f \right\|_n^2 \leq 18\lambda_n^{\mathcal{G}} \|\boldsymbol{\theta}^*\|_{\mathcal{G},w} + 24\mathcal{B}_{J_n}^2/\alpha' + 3\sigma^2 p/(n\alpha'')$, which is the first display of Propo-

sition 3. For the chain, Lemma S5.2(iii) gives $\lambda_n^{\mathcal{G}} w_G \leq \varrho \lambda_{n,1}^{\mathcal{G}}$ for every G , and summing with the weights $\|\boldsymbol{\theta}_G^*\|_2$ gives $\lambda_n^{\mathcal{G}} \|\boldsymbol{\theta}^*\|_{\mathcal{G},w} \leq \varrho \lambda_{n,1}^{\mathcal{G}} \|\boldsymbol{\theta}^*\|_{\mathcal{G},1}$, with $\varrho \leq \bar{\varrho}$; and $\|x\|_2 \leq \|x\|_1$ in each chunk gives $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} \leq \|\boldsymbol{\theta}^*\|_1$. The order of $\lambda_{n,1}^{\mathcal{G}}$ is (S6.1).

Part (i). With $s > 1/2$ and $2^{j^*+1} \geq b$, $\sum_{j>j^*} 2^{j(1/2-s)} \leq 2^{(j^*+1)(1/2-s)}/(1-2^{1/2-s}) \leq b_n^{1/2-s}/(1-2^{1/2-s})$, and Lemma S6.2 gives $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} \leq pqC_g\{(1-2^{-s'})^{-1} + b_n^{1/2-s-\vartheta_\pi}(1-2^{1/2-s})^{-1}\} = O(1)$; hence $\lambda_n^{\mathcal{G}} \|\boldsymbol{\theta}^*\|_{\mathcal{G},w} = O\{(b_n/n)^{1/2}\} = O\{(\log n/n)^{1/2}\}$. With $J_n \geq \log_2 n/(4s')$, $\mathcal{B}_{J_n}^2 \asymp 2^{-2J_n s'} \leq n^{-1/2}$, and p/n is of smaller order. With $s' > 1/4$ the choice $J_n = \lceil \log_2 n/(4s') \rceil$ has $2^{J_n} < 2n^{1/(4s')}$ and satisfies the conditions of the statement.

Part (ii). With $s < 1/2$, $\sum_{j^*<j<J_n} 2^{j(1/2-s)} \leq 2^{J_n(1/2-s)}/(2^{1/2-s}-1)$, so that $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} = O\{1 + b_n^{-\vartheta_\pi} 2^{J_n(1/2-s)}\}$ and, since $1/2 - \vartheta_\pi = (1/\pi - 1/2)_+$,

$$\lambda_n^{\mathcal{G}} \|\boldsymbol{\theta}^*\|_{\mathcal{G},w} \lesssim \left(\frac{b_n}{n}\right)^{1/2} + n^{-1/2} b_n^{(1/\pi-1/2)_+} 2^{J_n(1/2-s)}.$$

Equating the second term with the bias $2^{-2J_n s'}$ gives $2^{J_n(1/2-s+2s')} \asymp n^{1/2} b_n^{-(1/\pi-1/2)_+}$,

which is the choice of the statement, with common value $2^{-2J_n s'} \asymp \{b_n^{(2/\pi-1)_+}/n\}^{2s'/(1-2s+4s')}$.

The term $(b_n/n)^{1/2}$ and p/n are of smaller order, because $2s'/(1-2s+4s') < 1/2$ is equivalent to $s < 1/2$. The conditions of the statement hold because $s' > s/2$ gives $1/(1-2s+4s') < 1$, and $2^{J_n} \geq b_n$ for n large. When $\pi \geq 2$, $s' = s$, $(2/\pi - 1)_+ = 0$ and $1 - 2s + 4s = 2s + 1$. $\square$

Proof of Theorem 3

By the definition of J_n , $n^{1/(2s'+1)} \leq 2^{J_n} < 2n^{1/(2s'+1)} \leq 2n$, so the setting of this section holds, and $pq2^{J_n} \log(pq2^{J_n})/n \asymp n^{1/(2s'+1)-1} \log n = n^{-2s'/(2s'+1)} \log n \rightarrow 0$, which is the requirement of Proposition 2(ii); by Lemma S6.1, $P(\mathcal{A}_n) \rightarrow 1$. Fix $\epsilon > 0$, take $\alpha = \alpha' = \alpha'' = \epsilon/4$, and intersect $\mathcal{A}_n$, the event $\mathcal{E}$ of Lemma S5.2(i), $\{\|\mathbf{b}\|_n^2 \leq \mathcal{B}_{J_n}^2/\alpha'\}$ (Markov's inequality, since $E\|\mathbf{b}\|_n^2 = \|f - f_{J_n}\|_{L_2(P)}^2 \leq \mathcal{B}_{J_n}^2$ by Lemma 2) and $\{\|\Pi_{\mathbf{A}}\boldsymbol{\epsilon}\|_n^2 \leq \sigma^2 p/(n\alpha'')\}$ (Markov's inequality and Theorem 2(iii)); for n large the intersection has probability at least $1 - \epsilon$. On it $\mathcal{T}_{\mathcal{G},w}$ holds, and (S5.7) with $\bar{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ gives

$$\left\| \hat{f} - f \right\|_n^2 \leq \frac{192(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_0(\boldsymbol{\theta}^*))}{\tilde{\gamma}} + 60 \|\mathbf{b}\|_n^2 + 3 \|\Pi_{\mathbf{A}}\boldsymbol{\epsilon}\|_n^2.$$

By Lemma S5.2(iii), Lemma S6.1 and Proposition S5.1(ii), with $\tilde{\gamma} > \gamma/2$,

$$\frac{192(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_n)}{\tilde{\gamma}} \leq \frac{384\bar{\varrho}^2(\lambda_{n,1}^{\mathcal{G}})^2 |\mathcal{G}_n|}{\gamma} \leq \frac{384\bar{\varrho}^2 C_{\lambda} \sigma^2 pq(b_n + 2^{J_n})}{\gamma n},$$

which is of order $n^{-2s'/(2s'+1)}$, because $2^{J_n} \asymp n^{1/(2s'+1)}$ dominates $b_n \asymp \log n$. The bias term is $60\mathcal{B}_{J_n}^2/\alpha' \asymp 2^{-2J_n s'} \leq n^{-2s'/(2s'+1)}$, by the lower bound on 2^{J_n} , and the last term is $O(1/n)$, of strictly smaller order. Hence $P\{\left\| \hat{f} - f \right\|_n^2 > M_{\epsilon} n^{-2s'/(2s'+1)}\} < \epsilon$ for a constant M_{ϵ} . $\square$

Proof of Corollary 1

By the orthonormality of the wavelets of level below J_n in $L_2[0, 1]$ and Lemma 2, $\sum_{\ell, m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0, 1]}^2 \leq 2 \left\| \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \right\|_2^2 + 2pqC_g^2(1 - 2^{-2s'})^{-1}2^{-2J_ns'}$. On the event of the proof of Theorem 3, Theorem 2(ii) with $\bar{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ gives $\left\| \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \right\|_2^2 \leq 64(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_n) / \tilde{\gamma}^2 + 16 \|\mathbf{b}\|_n^2 / \tilde{\gamma}$, with $\tilde{\gamma} > \gamma/2$: the two terms have the order of those of Theorem 3, with γ^{-2} in place of γ^{-1} . For the levels, Lemma S5.1 with $\bar{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ gives $\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c}) = \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}) - \Pi_{\mathbf{A}}\mathbf{B}v$, so that $\|\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c})\|_n \leq \|\mathbf{b}\|_n + \|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n + \Lambda^{1/2} \|v\|_2$, and $\lambda_{\min}(\mathbf{A}^\top \mathbf{A} / n) > \gamma/2$ gives $\|\widehat{\mathbf{c}} - \mathbf{c}\|_2^2 \leq 2 \|\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c})\|_n^2 / \gamma$. Finally the components of a functional coefficient depend on distinct coordinates of u and the basis is orthonormal, so that $\|\widehat{\beta}_\ell - \beta_\ell\|_{L_2([0, 1]^q)} \leq |\widehat{c}_\ell - c_\ell| + \sum_m \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0, 1]}$, with q fixed. $\square$

Proof of Theorem 1

The window. Since $s' \leq s$, $s / \{(2s + 1)s'\} \geq 1 / (2s + 1)$, and the window is nonempty if and only if $s / \{(2s + 1)s'\} < 1$, that is $s' > s / (2s + 1)$. With $c < 1$, $2^{J_n} \leq 2n^c \leq 2n$, so the setting of this section holds; the requirement of Proposition 2(ii) is $pq2^{J_n} \log(pq2^{J_n}) / n \asymp n^{c-1} \log n \rightarrow 0$, and $P(\mathcal{A}_n) \rightarrow 1$ by Lemma S6.1.

The rate. Fix $\epsilon > 0$ and the event of the proof of Theorem 3. On it, Corollary S5.1 with $\Lambda' = \Lambda$ gives $\left\| \widehat{f} - f \right\|_n^2 \leq 120\Lambda R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta) + 120 \|\mathbf{b}\|_n^2 +$

$3 \|\Pi_{\mathbf{A}} \boldsymbol{\varepsilon}\|_n^2$, and, by Lemma S5.2(iii), (S6.1) and $\tilde{\gamma} > \gamma/2$,

$$\eta = \frac{1.6 \max_G (\lambda_n^{\mathcal{G}} w_G)^2}{\Lambda \tilde{\gamma}} \leq \frac{3.2 \bar{\varrho}^2 (\lambda_{n,1}^{\mathcal{G}})^2}{\Lambda \gamma} \leq \bar{\eta}_n := \kappa \frac{b_n}{n}, \quad \kappa = \frac{3.2 \bar{\varrho}^2 C_{\lambda} \sigma^2}{\Lambda \gamma}.$$

Since $R_{\mathcal{G}}$ is nondecreasing in η , Lemma 3 with $b = b_n$ gives

$$R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta) \leq R_{\mathcal{G}}(\boldsymbol{\theta}^*; \bar{\eta}_n) \leq pq \left\{ A_{s,\pi}^{\mathcal{G}} C_g^{\tau} \kappa^{2s/(2s+1)} n^{-2s/(2s+1)} b_n^{(2/\pi-1)_+/(2s+1) + \kappa \frac{b_n}{n}} \right\},$$

with $b_n \asymp \log n$; the second term is of smaller order because $2s/(2s+1) < 1$.

The bias is $O(2^{-2J_n s'}) = O(n^{-2cs'})$, and $2cs' \geq 2s/(2s+1)$ by the window; the term $\sigma^2 p/n$ is of smaller order. For the components, the second bound of Corollary S5.1 and Lemma 2 give the same rate, as in the proof of Corollary 1.

Two readings. The constant grows with the weights at most as $\bar{\varrho}^{4s/(2s+1)}$ in the leading term, through $\kappa^{2s/(2s+1)}$. The gain over the rate $(\log n/n)^{2s/(2s+1)}$ of the lasso (Theorem S8.3) is the factor $(\log n)^{2s'/(2s+1)}$, because $2s/(2s+1) - (2/\pi - 1)_+/(2s+1) = 2s'/(2s+1)$: when $\pi < 2$, $2s - 2/\pi + 1 = 2(s - 1/\pi + 1/2) = 2s'$, and when $\pi \geq 2$, $s' = s$. The logarithm of the union bound over the columns is absorbed by the size of the chunks, which is why they must have at least of the order of $\log |\mathcal{G}_n|$ coefficients. $\square$

Remark S6.1 (The lower bound of Klopp and Pensky with one modulator). For $q = 1$, $\mathbf{X}$ independent of U_1 and Gaussian errors, Klopp and Pensky (2015, Theorem 1, eqs. 3.6 and 3.7), with the number of covariates

fixed, give the minimax lower bound $n^{-2r/(2r+1)}$ over their class (A3) with the periodized wavelet basis, which their condition (A1)(b) admits. That class, a weighted ℓ_ν ball $\sum_k |a_k|^\nu (k+1)^{\nu(r+1/2-1/\nu)} \leq C_a^\nu$ over all the coefficients, implies Assumption 5 with $(s, \pi) = (r, \nu)$ and $C_g = C_a$, level by level, since $k+1 \geq 2^j$ at level j ; so the lower bound holds over the class of Assumption 5, and when $\pi \geq 2$ Theorem 1 is minimax optimal up to a constant in that case. When $\pi < 2$ the factor $(\log n)^{(2/\pi-1)/(2s+1)}$ remains, the same as in eqs. (3.19) and (3.20) of Klopp and Pensky with the number of covariates fixed, and the same as in the rate of block thresholding in the sequence model (Cai, 1999, Theorem 4, eq. 5.4), which holds for $s \geq 1/\pi$; the condition $s' > s/(2s+1)$ of Theorem 1 is implied by $s \geq 1/\pi$, since then $s' \geq 1/2 > s/(2s+1)$, and is strictly weaker. That this factor is necessary for the estimator is proved neither here nor by Klopp and Pensky, and for $q \geq 2$ no lower bound is available.

S7 Recovery of the block structure by thresholding

Throughout this section $\nu_{\ell m}$, $\widehat{\nu}_{\ell m}$, $\mathcal{S}$, $\widehat{\mathcal{S}}(t)$, $\widehat{\rho}_n^g$ and $\nu_{\min, n}$ are as in Section 3.6 of the paper, $\nu_{\min} = \min_{(\ell, m) \in \mathcal{S}} \nu_{\ell m}$ for a fixed sample size, and

$$\Delta = \max_{\ell, m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$$

is the largest error over the blocks. The first statement is deterministic and knows nothing about the `lasso` estimator.

Lemma S7.1 (Thresholding under a uniform block error). *Let $\{g_{\ell m}\}$ and $\{\widehat{g}_{\ell m}\}$ be any two families of functions in $L_2[0, 1]$ indexed by the blocks, with $\nu_{\ell m}$, $\widehat{\nu}_{\ell m}$, $\mathcal{S}$, $\widehat{\mathcal{S}}(t)$, Δ and $\nu_{\min}$ defined from them as above, and let $\bar{\Delta} \geq \Delta$. Then, for every $t \geq 0$,*

$$\{(\ell, m) : \nu_{\ell m} > t + \bar{\Delta}\} \subseteq \widehat{\mathcal{S}}(t) \subseteq \{(\ell, m) : \nu_{\ell m} > t - \bar{\Delta}\}. \quad (\text{S7.1})$$

In particular: (i) if $t \geq \bar{\Delta}$, then $\widehat{\mathcal{S}}(t) \subseteq \mathcal{S}$, with no use of $\nu_{\min}$; (ii) if $t < \nu_{\min} - \bar{\Delta}$, then $\mathcal{S} \subseteq \widehat{\mathcal{S}}(t)$; and (iii) if $\bar{\Delta} \leq t < \nu_{\min} - \bar{\Delta}$, then $\widehat{\mathcal{S}}(t) = \mathcal{S}$, and such a t exists if and only if $\nu_{\min} > 2\bar{\Delta}$.

Proof. By the triangle inequality in $L_2[0, 1]$, for every block, $|\widehat{\nu}_{\ell m} - \nu_{\ell m}| \leq \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]} \leq \Delta \leq \bar{\Delta}$. If $\nu_{\ell m} > t + \bar{\Delta}$, then $\widehat{\nu}_{\ell m} \geq \nu_{\ell m} - \bar{\Delta} > t$, which is the left inclusion of (S7.1); if $\widehat{\nu}_{\ell m} > t$, then $\nu_{\ell m} \geq \widehat{\nu}_{\ell m} - \bar{\Delta} > t - \bar{\Delta}$, which is the right one. For (i), $t - \bar{\Delta} \geq 0$ and the right inclusion give $\widehat{\mathcal{S}}(t) \subseteq \{\nu_{\ell m} > 0\} = \mathcal{S}$. For (ii), every block of $\mathcal{S}$ has $\nu_{\ell m} \geq \nu_{\min} > t + \bar{\Delta}$, and the left inclusion applies. Part (iii) is the conjunction of the two, and $[\bar{\Delta}, \nu_{\min} - \bar{\Delta}]$ is nonempty exactly when $\bar{\Delta} < \nu_{\min} - \bar{\Delta}$. $\square$

Remark S7.1 (The window is exact). The condition $\nu_{\min} > 2\bar{\Delta}$ is not an artifact of the proof. Take two blocks, one active with $\nu_{11} = 2\bar{\Delta}$ and one

null with $\nu_{21} = 0$, and fitted norms $\widehat{\nu}_{11} = \widehat{\nu}_{21} = \bar{\Delta}$, which is compatible with $\Delta \leq \bar{\Delta}$: the two statistics tie and no threshold separates them. This says that the argument does not extend below $\nu_{\min} = 2\bar{\Delta}$, not that no procedure does, which would be a lower bound and is not proved here.

Lemma S7.2 (The risk of the thresholded estimator). *Let $\{g_{\ell m}\}$ and $\{\widehat{g}_{\ell m}\}$ be any two families of functions in $L_2[0, 1]$ indexed by the blocks and $t \geq 0$, with $\nu_{\ell m}$, $\widehat{\nu}_{\ell m}$, $\mathcal{S}$ and $\widehat{\mathcal{S}}(t)$ as above, $\Delta_{\ell m} = \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$, and $\widehat{g}_{\ell m}^{(t)} = \widehat{g}_{\ell m} \mathbf{1}\{(\ell, m) \in \widehat{\mathcal{S}}(t)\}$.*

(i) *With no assumption,*

$$\sum_{\ell, m} \|\widehat{g}_{\ell m}^{(t)} - g_{\ell m}\|_{L_2[0,1]}^2 = \sum_{(\ell, m) \in \widehat{\mathcal{S}}(t)} \Delta_{\ell m}^2 + \sum_{(\ell, m) \in \mathcal{S} \setminus \widehat{\mathcal{S}}(t)} \nu_{\ell m}^2,$$

and every $(\ell, m) \in \mathcal{S} \setminus \widehat{\mathcal{S}}(t)$ has $\nu_{\ell m} \leq t + \Delta_{\ell m}$; hence the left side is

$$\text{at most } 2 \sum_{\ell, m} \Delta_{\ell m}^2 + 2t^2 |\mathcal{S} \setminus \widehat{\mathcal{S}}(t)| \leq 2 \sum_{\ell, m} \Delta_{\ell m}^2 + 2t^2 |\mathcal{S}|.$$

(ii) *If $\mathcal{S} \subseteq \widehat{\mathcal{S}}(t)$, the left side of (i) is $\sum_{\widehat{\mathcal{S}}(t)} \Delta_{\ell m}^2 \leq \sum_{\ell, m} \Delta_{\ell m}^2$; if $\widehat{\mathcal{S}}(t) = \mathcal{S}$, it is $\sum_{\mathcal{S}} \Delta_{\ell m}^2$, and the null blocks no longer contribute.*

(iii) *Let $\lambda_{\min}(\widehat{\Sigma}) > 0$, $\beta^* = (\mathbf{c}, \boldsymbol{\theta}^*)$, $f_J = \mathbf{Z}\beta^*$, $\widehat{\beta} = (\widehat{\mathbf{c}}, \widehat{\boldsymbol{\theta}})$ any fit, $\widehat{\mathcal{S}}(t) = \{(\ell, m) : \|\widehat{\boldsymbol{\theta}}_{\ell m, \cdot}\|_2 > t\}$, $\mathcal{S}_J = \{(\ell, m) : \boldsymbol{\theta}_{\ell m, \cdot}^* \neq 0\}$, $\widehat{\boldsymbol{\theta}}^{(t)}$ the vector $\widehat{\boldsymbol{\theta}}$ with the blocks outside $\widehat{\mathcal{S}}(t)$ set to zero, and $\widehat{f}^{(t)} = \mathbf{A}\widehat{\mathbf{c}} + \mathbf{B}\widehat{\boldsymbol{\theta}}^{(t)}$. Then*

$$\|\widehat{f}^{(t)} - f_J\|_n^2 \leq \frac{2\lambda_{\max}(\widehat{\Sigma})}{\lambda_{\min}(\widehat{\Sigma})} \|\widehat{f} - f_J\|_n^2 + 2\lambda_{\max}(\widehat{\Sigma}) t^2 |\mathcal{S}_J \setminus \widehat{\mathcal{S}}(t)|,$$

and, if $\mathcal{S}_J \subseteq \widehat{\mathcal{S}}(t)$, $\left\| \widehat{f}^{(t)} - f_J \right\|_n^2 \leq \{\lambda_{\max}(\widehat{\Sigma})/\lambda_{\min}(\widehat{\Sigma})\} \left\| \widehat{f} - f_J \right\|_n^2$.

Proof. (i). If $(\ell, m) \in \widehat{\mathcal{S}}(t)$, then $\widehat{g}_{\ell m}^{(t)} = \widehat{g}_{\ell m}$ and the error of the block is $\Delta_{\ell m}^2$; otherwise $\widehat{g}_{\ell m}^{(t)} = 0$ and the error is $\nu_{\ell m}^2$, which vanishes outside $\mathcal{S}$. Summing gives the identity. For $(\ell, m) \notin \widehat{\mathcal{S}}(t)$, $\widehat{\nu}_{\ell m} \leq t$, and the triangle inequality gives $\nu_{\ell m} \leq \widehat{\nu}_{\ell m} + \Delta_{\ell m} \leq t + \Delta_{\ell m}$, hence $\nu_{\ell m}^2 \leq 2t^2 + 2\Delta_{\ell m}^2$; summing over $\mathcal{S} \setminus \widehat{\mathcal{S}}(t)$ and adding the sum over $\widehat{\mathcal{S}}(t)$, a disjoint set, gives the bound. (ii). If $\mathcal{S} \subseteq \widehat{\mathcal{S}}(t)$, the second sum of (i) is empty. (iii). The proof of (i) and (ii) uses only the triangle inequality in a norm; applied to the vectors $\widehat{\boldsymbol{\theta}}_{\ell m, \cdot}$ and $\boldsymbol{\theta}_{\ell m, \cdot}^*$ of $\mathbb{R}^{N_J}$, with $\mathcal{S}_J$ in place of $\mathcal{S}$, it gives $\left\| \widehat{\boldsymbol{\theta}}^{(t)} - \boldsymbol{\theta}^* \right\|_2^2 \leq 2 \left\| \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \right\|_2^2 + 2t^2 |\mathcal{S}_J \setminus \widehat{\mathcal{S}}(t)|$, and $\left\| \widehat{\boldsymbol{\theta}}^{(t)} - \boldsymbol{\theta}^* \right\|_2^2 \leq \left\| \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \right\|_2^2$ when $\mathcal{S}_J \subseteq \widehat{\mathcal{S}}(t)$. Since $\widehat{\mathbf{c}}$ does not change, the same bounds hold for $\widehat{\boldsymbol{\beta}}^{(t)} = (\widehat{\mathbf{c}}, \widehat{\boldsymbol{\theta}}^{(t)})$ against $\boldsymbol{\beta}^*$. Finally $\left\| \widehat{f}^{(t)} - f_J \right\|_n^2 = (\widehat{\boldsymbol{\beta}}^{(t)} - \boldsymbol{\beta}^*)^\top \widehat{\Sigma} (\widehat{\boldsymbol{\beta}}^{(t)} - \boldsymbol{\beta}^*) \leq \lambda_{\max}(\widehat{\Sigma}) \left\| \widehat{\boldsymbol{\beta}}^{(t)} - \boldsymbol{\beta}^* \right\|_2^2$ and $\left\| \widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^* \right\|_2^2 \leq \left\| \mathbf{Z}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*) \right\|_n^2 / \lambda_{\min}(\widehat{\Sigma}) = \left\| \widehat{f} - f_J \right\|_n^2 / \lambda_{\min}(\widehat{\Sigma})$. By the orthonormality of the basis, $\left\| \widehat{\boldsymbol{\theta}}_{\ell m, \cdot} \right\|_2 = \widehat{\nu}_{\ell m}$, so that the $\widehat{\mathcal{S}}(t)$ of (iii) is that of (i). $\square$

The factor 2 in (i) is attained: an active block with $\widehat{g}_{\ell m} = t g_{\ell m} / \nu_{\ell m}$ and $\nu_{\ell m} = 2t$ has $\widehat{\nu}_{\ell m} = t$, is set to zero, and pays $\nu_{\ell m}^2 = 4t^2 = 2\Delta_{\ell m}^2 + 2t^2$. In (iii), the passage between the norm of the coefficients and the prediction norm uses the smallest eigenvalue of $\widehat{\Sigma}$ to go from prediction to coefficients

and the largest to come back, and the factor is the condition number of $\widehat{\Sigma}$, which on $\mathcal{E}_\Sigma$ is smaller than $2\Lambda/\gamma$.

[k = 4: the proof of Corollary 2 for the lasso, which stood here, is moved to Section S8, unchanged except for the numbering of the results it cites.]

Proof of Corollary 2

A *nonasymptotic bound on Δ* . Work at $J = J_n$ and $\lambda = \lambda_n^{\mathcal{G}}$, and intersect $\mathcal{A}_n$, the event $\mathcal{E}$ of Lemma S5.2(i) and $\{\|\mathbf{b}\|_n^2 \leq \mathcal{B}_{J_n}^2/\alpha'\}$; the intersection has probability at least $1 - \alpha - \alpha' - \mathbb{P}(\mathcal{A}_n^c)$, by Lemma S5.2, Markov's inequality and Lemma 2. On $\mathcal{A}_n$, $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$ and $\tilde{\gamma} > \gamma/2$ (Lemma S6.1), and Theorem 2(ii) with the comparator β^* and $\mathcal{W}(\mathcal{G}_0(\theta^*)) \leq \mathcal{W}(\mathcal{G}_n)$ gives, with $v = \widehat{\theta} - \theta^*$,

$$\|v\|_2^2 \leq \frac{64(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_n)}{\tilde{\gamma}^2} + \frac{16 \|\mathbf{b}\|_n^2}{\tilde{\gamma}} \leq \frac{256(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_n)}{\gamma^2} + \frac{32\mathcal{B}_{J_n}^2}{\alpha'\gamma}.$$

For a fixed block, the orthonormality of the wavelets of level below J_n gives $\|\widehat{g}_{\ell m} - \Pi_J g_{\ell m}\|_{L_2[0,1]} = \|v_{\ell m, \cdot}\|_2 \leq \|v\|_2$, and the first display of Lemma 2 bounds $\|\Pi_J g_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2$ by $C_g^2(1 - 2^{-2s'})^{-1}2^{-2J_n s'}$. By $(x + y)^2 \leq 2x^2 + 2y^2$,

$$\Delta^2 \leq (\bar{\Delta}_n^{\mathcal{G}})^2 = \frac{512(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_n)}{\gamma^2} + \frac{64\mathcal{B}_{J_n}^2}{\alpha'\gamma} + \frac{2C_g^2}{1 - 2^{-2s'}} 2^{-2J_n s'}, \quad (\text{S7.2})$$

with $(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_n) \leq \varrho^2 (\lambda_{n,1}^{\mathcal{G}})^2 |\mathcal{G}_n|$ by Lemma S5.2(iii). The residual probability $P(\mathcal{A}_n^c)$ tends to zero by Lemma S6.1 and does not depend on the components; it is the ω_n of Corollary 2(iii), and by Lemma S7.1(iii) this paragraph is the proof of that part. The bound $\bar{\Delta}_n^{\mathcal{G}}$ depends on the components only through the constants of Assumption 5, so everything above holds uniformly over the class, which is what allows the target to vary with n in Assumption 6.

The order of $\bar{\Delta}_n^{\mathcal{G}}$. By the proof of Theorem 3, $(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_n) \leq \bar{\varrho}^2 C_{\lambda} \sigma^2 p q (b_n + 2^{J_n})/n \asymp n^{-2s'/(2s'+1)}$, and $2^{-2J_n s'} \leq n^{-2s'/(2s'+1)}$ by the definition of J_n . Hence $\bar{\Delta}_n^{\mathcal{G}} \leq c(\alpha, \alpha') \rho_n^{\mathcal{G}}$ with $c(\alpha, \alpha') < \infty$. Given $\epsilon > 0$, take $\alpha = \alpha' = \epsilon/4$ and n large enough that $P(\mathcal{A}_n^c) < \epsilon/2$; then $P\{\Delta > c(\epsilon/4, \epsilon/4) \rho_n^{\mathcal{G}}\} < \epsilon$, that is $\Delta = O_p(\rho_n^{\mathcal{G}})$, uniformly over the class.

Part (i). If $t_n/\rho_n^{\mathcal{G}} \rightarrow \infty$, then $P(\Delta \leq t_n) \rightarrow 1$, and on that event Lemma S7.1(i) with $\bar{\Delta} = \Delta$ gives $\widehat{\mathcal{S}}(t_n) \subseteq \mathcal{S}$. No property of $\nu_{\min,n}$ has been used.

Part (ii). If $\mathcal{S}$ is empty, part (ii) is part (i). Otherwise, the condition $\limsup_n t_n/\nu_{\min,n} < 1$ gives $\eta > 0$ and n_0 with $\nu_{\min,n} - t_n \geq \eta \nu_{\min,n}$ for $n \geq n_0$, and Assumption 6 gives $(\nu_{\min,n} - t_n)/\rho_n^{\mathcal{G}} \geq \eta \nu_{\min,n}/\rho_n^{\mathcal{G}} \rightarrow \infty$. Hence $P(\Delta \leq t_n) \rightarrow 1$ and $P(\Delta < \nu_{\min,n} - t_n) \rightarrow 1$, and on the intersection of the two events Lemma S7.1(iii) with $\bar{\Delta} = \Delta$ gives $\widehat{\mathcal{S}}(t_n) = \mathcal{S}$.

For the existence, $t_n = (\rho_n^{\mathcal{G}} \nu_{\min,n})^{1/2}$ has $t_n/\rho_n^{\mathcal{G}} = (\nu_{\min,n}/\rho_n^{\mathcal{G}})^{1/2} \rightarrow \infty$ and $t_n/\nu_{\min,n} = (\rho_n^{\mathcal{G}}/\nu_{\min,n})^{1/2} \rightarrow 0$. $\square$

At the resolution of Theorem 1. Theorem 1 gives $\sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 = O_p\{(\tilde{\rho}_n^{\mathcal{G}})^2\}$, with $\tilde{\rho}_n^{\mathcal{G}}$ as in Section 3.6 of the paper, hence $\Delta = O_p(\tilde{\rho}_n^{\mathcal{G}})$, and the bound in its proof depends on the components only through the constants of Assumption 5 and of the design, so that this holds uniformly over the class. The proofs of parts (i) and (ii) above use nothing about Δ beyond $\Delta = O_p(\rho_n^{\mathcal{G}})$ uniformly over the class, together with Lemma S7.1 and Assumption 6, and they go through verbatim with $\tilde{\rho}_n^{\mathcal{G}}$ in place of $\rho_n^{\mathcal{G}}$.

Proof of Corollary 3

Write r_n for the rate of the statement and $\Delta_{\ell m}$ as in Lemma S7.2. For the fit of Theorem 3, $\|\widehat{f} - f\|_n^2 = O_p(r_n^2)$ and $\sum_{\ell,m} \Delta_{\ell m}^2 = O_p(r_n^2)$ by Theorem 3 and Corollary 1; for the lasso, by Theorem S8.2 and Corollary S8.1. In both cases $P(\mathcal{E}_{\Sigma}) \rightarrow 1$, by the proofs of Theorems 3 and S8.2, and $\|\mathbf{b}\|_n^2 = O_p(\mathcal{B}_{J_n}^2) = O_p(r_n^2)$, by the choice of J_n .

Part (i). The bound for the components is Lemma S7.2(i) with $|\mathcal{S}| \leq pq$. For prediction, $\|\widehat{f}^{(t)} - f\|_n^2 \leq 2\|\widehat{f}^{(t)} - f_{J_n}\|_n^2 + 2\|\mathbf{b}\|_n^2$, and on $\mathcal{E}_{\Sigma}$, $\lambda_{\max}(\widehat{\Sigma}) < \Lambda$ and $\lambda_{\min}(\widehat{\Sigma}) > \gamma/2$, so that Lemma S7.2(iii) gives, for every t ,

$$\|\widehat{f}^{(t)} - f\|_n^2 \leq \frac{8\Lambda}{\gamma} \|\widehat{f} - f_{J_n}\|_n^2 + 4\Lambda t^2 |\mathcal{S}_{J_n} \setminus \widehat{\mathcal{S}}(t)| + 2\|\mathbf{b}\|_n^2,$$

with $\left\|\widehat{f} - f_{J_n}\right\|_n^2 \leq 2 \left\|\widehat{f} - f\right\|_n^2 + 2 \|\mathbf{b}\|_n^2 = O_p(r_n^2)$.

Part (ii). By Lemma S7.1(ii), $\mathcal{S} \subseteq \widehat{\mathcal{S}}(t_n)$ whenever $t_n < \nu_{\min,n} - \Delta$.

With $\limsup_n t_n/\nu_{\min,n} < 1$ there are $\eta > 0$ and n_0 with $\nu_{\min,n} - t_n \geq \eta\nu_{\min,n}$ for $n \geq n_0$, and since $\Delta^2 \leq \sum_{\ell,m} \Delta_{\ell m}^2 = O_p(r_n^2)$ and $\nu_{\min,n}/r_n \rightarrow \infty$, $P(\Delta < \nu_{\min,n} - t_n) \rightarrow 1$. On that event Lemma S7.2(ii) holds and, since $\mathcal{S}_{J_n} \subseteq \mathcal{S}$, so does the second case of Lemma S7.2(iii), which on $\mathcal{E}_\Sigma$ gives $\left\|\widehat{f}^{(t_n)} - f\right\|_n^2 \leq (4\Lambda/\gamma) \left\|\widehat{f} - f_{J_n}\right\|_n^2 + 2 \|\mathbf{b}\|_n^2$. If moreover $t_n/r_n \rightarrow \infty$, part (ii) of Corollary 2, or of Corollary S8.2 for the lasso, gives $P\{\widehat{\mathcal{S}}(t_n) = \mathcal{S}\} \rightarrow 1$, and Lemma S7.2(ii) concludes. $\square$

S8 The coordinatewise lasso

[k = 4: the statements of this section stood in Sections 3.2 to 3.6 of the paper, and their proofs in Sections S5 to S7 of this supplement.]

This section gives the theory of the estimator obtained by replacing $\|\boldsymbol{\theta}\|_{\mathcal{G},w}$ by $\|\boldsymbol{\theta}\|_1$ in equation (2.5) of the paper, the lasso of Tibshirani (1996) with the levels left unpenalized, which the software offers as an option. Assumptions 1 to 5, Lemmas 1 and 2, Proposition 2 and Lemmas S5.1, S5.3 and S5.4 are used as they stand, with the notation S_0 , s_0 and v of

Section S1. The penalty level is

$$\lambda_n = 2\sigma B_X \sqrt{2C_U} \sqrt{\frac{2\log(2d_n/\alpha)}{n}} \asymp \sigma B_X \sqrt{C_U} \sqrt{\frac{J_n}{n}}, \quad (\text{S8.1})$$

with d_n the number of penalized columns at resolution J_n .

Statements

Theorem S8.1 (Oracle inequality for the lasso). *Let Assumption 1 hold, suppose $\mathbf{A}$ has rank p , and take $\lambda \geq 2\lambda_0$ with $\lambda_0 = \sigma \hat{\sigma}_{\max}\{2\log(2d/\alpha)/n\}^{1/2}$.*

On the event $\mathcal{T} = \{\|\tilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n\|_\infty \leq \lambda/2\}$, which has probability at least $1 - \alpha$:

(i) *with no condition on the design, $\|\tilde{\mathbf{B}}v\|_n^2 \leq 6\lambda \|\boldsymbol{\theta}^*\|_1 + 4\|\mathbf{b}\|_n^2$;*

(ii) *if $\tilde{\gamma} > 0$,*

$$\|\tilde{\mathbf{B}}v\|_n^2 \leq \frac{64\lambda^2 s_0}{\tilde{\gamma}} + 16\|\mathbf{b}\|_n^2, \quad \|v\|_2^2 \leq \frac{64\lambda^2 s_0}{\tilde{\gamma}^2} + \frac{16\|\mathbf{b}\|_n^2}{\tilde{\gamma}};$$

(iii) *in either regime, $\|\hat{f} - f\|_n \leq \|\tilde{\mathbf{B}}v\|_n + 2\|\mathbf{b}\|_n + \|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n$, and $\mathbb{E}(\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \mid$*

$$\mathbf{X}, \mathbf{U}) \leq \sigma^2 p/n.$$

Theorem S8.2 (Rate of linear approximation for the lasso). *Let Assumptions 1 to 5 hold with p, q and the constants fixed, and take*

$$J_n = \min\{J \in \mathbb{N} : 2^J \geq (n/\log n)^{1/(2s'+1)}\}, \quad (\text{S8.2})$$

so that $J_n = \log_2 n/(2s'+1) + O(\log \log n)$. Then the requirement of Proposition 2(ii) holds along J_n , being itself of the exact order $(\log n/n)^{2s'/(2s'+1)}$,

and, with λ_n as in (S8.1),

$$\left\| \widehat{f} - f \right\|_n^2 = O_p \left\{ \left(\frac{\log n}{n} \right)^{2s'/(2s'+1)} \right\}.$$

At this resolution the estimation term and the bias term are of the same order, and the term $\sigma^2 p/n$ of Theorem S8.1(iii) is of strictly smaller order.

Corollary S8.1 (Components and coefficients for the lasso). *In the setting of Theorem S8.2, and with r_n denoting the rate of that theorem,*

$$\sum_{\ell, m} \left\| \widehat{g}_{\ell m} - g_{\ell m} \right\|_{L_2[0,1]}^2 = O_p(r_n), \quad \left\| \widehat{\beta}_\ell - \beta_\ell \right\|_{L_2([0,1]^q)} = O_p(r_n^{1/2})$$

for $\ell = 1, \dots, p$, and the same statements hold with the rate of Theorem S8.3 in place of r_n under the conditions of that theorem.

Proposition S8.1 (Slow rate for the lasso, no design condition). *Let Assumptions 1, 4 and 5 hold, with $\mathbf{A}$ of rank p and λ_n as in (S8.1). Then*

$$\left\| \widehat{f} - f \right\|_n^2 = O_p \{ \lambda_n \left\| \boldsymbol{\theta}^* \right\|_1 + \mathcal{B}_{J_n}^2 + p/n \}, \text{ where } \left\| \boldsymbol{\theta}^* \right\|_1 \leq pqC_g \sum_{j < J_n} 2^{j(1/2-s)}$$

does not depend on π . In particular, if $s > 1/2$ and $s' > 1/4$ then

$$\left\| \widehat{f} - f \right\|_n^2 = O_p \{ (J_n/n)^{1/2} \} \text{ for every } J_n \geq \log_2 n / (4s'); \text{ and if } s < 1/2 \text{ and}$$

$$s' > s/2, \text{ the choice } 2^{J_n} \asymp (n/J_n)^{1/(1-2s+4s')} \text{ gives } \left\| \widehat{f} - f \right\|_n^2 = O_p \{ (J_n/n)^{2s'/(1-2s+4s')} \},$$

an exponent that equals $2s/(2s+1)$ when $\pi \geq 2$.

For a vector $\boldsymbol{\theta}$ let $\theta_{(1)} \geq \theta_{(2)} \geq \dots$ be the nonincreasing rearrangement of the absolute values of its entries and let $\boldsymbol{\theta}^{(M)}$ retain the M largest of them.

Lemma S8.1 (Besov regularity implies compressibility). *Under Assumptions 4 and 5 with $\pi(s+1/2) > 1$, the oracle satisfies, for every J and every $k \geq 1$,*

$$\theta_{(k)}^* \leq C_\tau k^{-1/\tau}, \quad \tau = \frac{1}{s+1/2}, \quad C_\tau = C_g(pq A_{s,\pi})^{s+1/2}, \quad A_{s,\pi} = 1 + \frac{1}{1 - 2^{1-\pi(s+1/2)}},$$

with constants that do not depend on J and with $\tau \in (0, 2)$ for every $s > 0$.

*Consequently $\sum_{k>M} \theta_{(k)}^{*2} \leq \tau(2-\tau)^{-1} C_\tau^2 M^{1-2/\tau}$ for every $M \geq 1$.*

The inclusion of Besov balls in weak ℓ_τ balls is classical (Johnstone, 2019, Proposition 9.15); what the lemma adds is the constant C_τ , uniform in J , for the oracle of the product design, which is the form in which the proof of Theorem S8.3 uses it.

Theorem S8.3 (Rate of nonlinear approximation for the lasso). *Let Assumptions 1 to 5 hold with p, q and all the constants fixed as n grows, and set $\tau = (s+1/2)^{-1}$. Suppose $s' > s/(2s+1)$ and take $J_n = \lceil c \log_2 n \rceil$ with*

$$\max\left\{\frac{\tau}{2}, \frac{2-\tau}{4s'}\right\} \leq c < 1, \tag{S8.3}$$

a window that is nonempty precisely under that condition. Then, with λ_n as in (S8.1),

$$\|\widehat{f} - f\|_n^2 = O_p\left\{\left(\frac{\log n}{n}\right)^{2s/(2s+1)}\right\}, \quad \sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 = O_p\left\{\left(\frac{\log n}{n}\right)^{2s/(2s+1)}\right\}.$$

The logarithmic factor is the price of the single penalty level (S8.1), which takes a union bound over the d_n columns; Theorem 1 of the paper

removes the factor $(\log n)^{2s'/(2s+1)}$ of it. Write $\rho_n = (\log n/n)^{s'/(2s'+1)}$ for the square root of the rate of Theorem S8.2.

Corollary S8.2 (Block structure by thresholding the lasso). *Let the assumptions of Theorem S8.2 hold, with J_n and λ_n as there.*

- (i) (Screening.) *If the thresholds satisfy $t_n/\rho_n \rightarrow \infty$, then $\mathbb{P}\{\widehat{\mathcal{S}}(t_n) \subseteq \mathcal{S}\} \rightarrow 1$.*
- (ii) (Recovery.) *If, in addition, Assumption 6 holds with ρ_n in place of ρ_n^G and $\limsup_n t_n/\nu_{\min,n} < 1$, then $\mathbb{P}\{\widehat{\mathcal{S}}(t_n) = \mathcal{S}\} \rightarrow 1$. Such thresholds always exist; one is $t_n = (\rho_n \nu_{\min,n})^{1/2}$.*
- (iii) (A window at fixed n .) *With α the level in (S8.1), for every n and every $\alpha' \in (0, 1)$ there is an event, of probability at least $1 - \alpha - \alpha' - \omega_n$ with $\omega_n \rightarrow 0$ not depending on the components, on which $\widehat{\mathcal{S}}(t) = \mathcal{S}$ for every t with $\bar{\Delta}_n \leq t < \nu_{\min,n} - \bar{\Delta}_n$; here $\bar{\Delta}_n \leq c(\alpha, \alpha')\rho_n$ is the explicit bound (S8.8) on $\max_{(\ell,m)} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$.*

Parts (i) and (ii) hold also at the resolution of Theorem S8.3, with ρ_n replaced by $\tilde{\rho}_n = (\log n/n)^{s/(2s+1)}$, the square root of the rate of that theorem, both in the condition on t_n and in Assumption 6. Since $\tilde{\rho}_n \leq \rho_n$, with a gap in the exponent when $\pi < 2$, the separation required is weaker exactly where the components are spatially inhomogeneous.

Proofs

Lemma S8.2 (Calibration of the penalty). *Under Assumption 1, for every $\alpha \in (0, 1)$, $P(\|\tilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n\|_\infty > \lambda_0) \leq \alpha$ with $\lambda_0 = \sigma \hat{\sigma}_{\max} \{2 \log(2d/\alpha)/n\}^{1/2}$.*

Proof. Condition on the covariates, so that $\tilde{\mathbf{B}}$ and $\hat{\sigma}_{\max}$ are fixed and the errors are independent and sub-Gaussian with parameter σ . For a column a , $(\tilde{\mathbf{B}}^\top \boldsymbol{\varepsilon})_a/n$ is sub-Gaussian with parameter $\sigma \|\tilde{\mathbf{B}}^{(a)}\|_2/n$, and, $M_{\mathbf{A}}$ being an orthogonal projection, $\|\tilde{\mathbf{B}}^{(a)}\|_2 \leq \|\mathbf{B}^{(a)}\|_2 = \sqrt{n} (\hat{\Sigma})_{aa}^{1/2} \leq \sqrt{n} \hat{\sigma}_{\max}$. With $t = \lambda_0$ the sub-Gaussian tail bound is at most α/d per column; a union bound over the d columns and the tower property finish the proof. $\square$

Proof of Theorem S8.1

Step 1, the basic inequality. By Lemma S5.1, $\hat{\boldsymbol{\theta}}$ minimizes the residualized objective; comparing its value at $\hat{\boldsymbol{\theta}}$ with its value at $\boldsymbol{\theta}^*$, and using $M_{\mathbf{A}} \mathbf{A} = 0$, so that $M_{\mathbf{A}} \mathbf{Y} = \tilde{\mathbf{B}} \boldsymbol{\theta}^* + M_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon})$, gives, with $r = M_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon})$,

$$\left\| \tilde{\mathbf{B}} v \right\|_n^2 + 2\lambda \left\| \hat{\boldsymbol{\theta}} \right\|_1 \leq 2 \langle r, \tilde{\mathbf{B}} v \rangle_n + 2\lambda \left\| \boldsymbol{\theta}^* \right\|_1. \quad (\text{S8.4})$$

Step 2, the two noises. Since $M_{\mathbf{A}} \tilde{\mathbf{B}} = \tilde{\mathbf{B}}$ and $M_{\mathbf{A}}$ is symmetric, $\langle r, \tilde{\mathbf{B}} v \rangle_n = \langle \mathbf{b} + \boldsymbol{\varepsilon}, \tilde{\mathbf{B}} v \rangle_n$. The stochastic part is handled by duality: on $\mathcal{T}$, $2|\langle \boldsymbol{\varepsilon}, \tilde{\mathbf{B}} v \rangle_n| \leq 2 \left\| \tilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n \right\|_\infty \|v\|_1 \leq \lambda \|v\|_1$. The deterministic part is handled by the Cauchy-Schwarz and Young inequalities, and not by duality, because $\mathbf{b}$

is not small coordinatewise: $2|\langle \mathbf{b}, \tilde{\mathbf{B}}v \rangle_n| \leq 2\|\mathbf{b}\|_n \|\tilde{\mathbf{B}}v\|_n \leq \|\tilde{\mathbf{B}}v\|_n^2/2 + 2\|\mathbf{b}\|_n^2$. Substituting both in (S8.4),

$$\frac{1}{2} \|\tilde{\mathbf{B}}v\|_n^2 + 2\lambda \|\hat{\boldsymbol{\theta}}\|_1 \leq \lambda \|v\|_1 + 2\lambda \|\boldsymbol{\theta}^*\|_1 + 2\|\mathbf{b}\|_n^2. \quad (\text{S8.5})$$

Step 3, the slow rate. Bounding $\|v\|_1 \leq \|\hat{\boldsymbol{\theta}}\|_1 + \|\boldsymbol{\theta}^*\|_1$ in (S8.5) gives $\frac{1}{2} \|\tilde{\mathbf{B}}v\|_n^2 + \lambda \|\hat{\boldsymbol{\theta}}\|_1 \leq 3\lambda \|\boldsymbol{\theta}^*\|_1 + 2\|\mathbf{b}\|_n^2$, which is part (i). No property of $\hat{\Sigma}$ was used.

Step 4, the cone. Since $\boldsymbol{\theta}^*$ vanishes off S_0 , $\|\hat{\boldsymbol{\theta}}\|_1 \geq \|\boldsymbol{\theta}^*\|_1 - \|v_{S_0}\|_1 + \|v_{S_0^c}\|_1$ and $\|v\|_1 = \|v_{S_0}\|_1 + \|v_{S_0^c}\|_1$; substituting in (S8.5) and cancelling $2\lambda \|\boldsymbol{\theta}^*\|_1$,

$$\frac{1}{2} \|\tilde{\mathbf{B}}v\|_n^2 + \lambda \|v_{S_0^c}\|_1 \leq 3\lambda \|v_{S_0}\|_1 + 2\|\mathbf{b}\|_n^2. \quad (\text{S8.6})$$

Step 5, two cases. If $2\|\mathbf{b}\|_n^2 \leq \lambda \|v_{S_0}\|_1$, then (S8.6) gives $\frac{1}{2} \|\tilde{\mathbf{B}}v\|_n^2 + \lambda \|v_{S_0^c}\|_1 \leq 4\lambda \|v_{S_0}\|_1$; by the Cauchy-Schwarz inequality and (S5.2), $\|v_{S_0}\|_1 \leq \sqrt{s_0} \|v\|_2 \leq \sqrt{s_0} \|\tilde{\mathbf{B}}v\|_n / \sqrt{\tilde{\gamma}}$, so $\|\tilde{\mathbf{B}}v\|_n \leq 8\lambda \sqrt{s_0/\tilde{\gamma}}$ and $\|v\|_2^2 \leq \|\tilde{\mathbf{B}}v\|_n^2 / \tilde{\gamma} \leq 64\lambda^2 s_0 / \tilde{\gamma}^2$; the same display gives $\|v_{S_0^c}\|_1 \leq 4\lambda \|v_{S_0}\|_1$ and hence $\|v\|_1 \leq 5\lambda \|v_{S_0}\|_1 \leq 40\lambda s_0 / \tilde{\gamma}$. If instead $2\|\mathbf{b}\|_n^2 > \lambda \|v_{S_0}\|_1$, then (S8.6) gives $\frac{1}{2} \|\tilde{\mathbf{B}}v\|_n^2 < 8\|\mathbf{b}\|_n^2$, that is $\|\tilde{\mathbf{B}}v\|_n^2 < 16\|\mathbf{b}\|_n^2$, together with $\lambda \|v\|_1 < 10\|\mathbf{b}\|_n^2$ and $\|v\|_2^2 < 16\|\mathbf{b}\|_n^2 / \tilde{\gamma}$. Each quantity is bounded by the maximum of the two cases, hence by their sum, which is part (ii).

Step 6, prediction. By (S5.1), $\hat{f} - f = \tilde{\mathbf{B}}v + \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}) - \mathbf{b}$, and $\Pi_{\mathbf{A}}$

being a projection, $\|\Pi_{\mathbf{A}}\mathbf{b}\|_n \leq \|\mathbf{b}\|_n$, which gives the triangle inequality of part (iii). Conditionally on the covariates the errors are independent, centered and of variance at most σ^2 , so

$$\mathbb{E}(\|\Pi_{\mathbf{A}}\varepsilon\|_n^2 \mid \mathbf{X}, \mathbf{U}) = \frac{1}{n} \text{tr}\{\Pi_{\mathbf{A}} \text{Cov}(\varepsilon \mid \mathbf{X}, \mathbf{U})\} \leq \frac{\sigma^2}{n} \text{tr}(\Pi_{\mathbf{A}}) \leq \frac{\sigma^2 p}{n},$$

since the diagonal entries of $\Pi_{\mathbf{A}}$ are nonnegative and its trace is the rank of $\mathbf{A}$. □

Remark S8.1 (Why there is no cone condition). The usual treatment of unpenalized coordinates (Bühlmann and van de Geer, 2011, Section 6.9) places them inside the support and requires compatibility on the cone $\|\delta_{S^c}\|_1 \leq 3\|\delta_S\|_1$. Profiling removes them from the problem, and (S5.2) transfers the full smallest eigenvalue of $\widehat{\Sigma}$ to the residualized block, so that Proposition 2(ii) is used as it stands and the constant does not depend on S_0 . If only the restricted version of the design condition were available, the cone would return, and with the constant 4 rather than 3, since Step 5 produces $\|v_{S_0^c}\|_1 \leq 4\|v_{S_0}\|_1$: the bias consumes part of the slack.

Corollary S8.3 (The two rates with the design condition instantiated).

Add Assumptions 2 to 5 and suppose $2^J \log d/n \rightarrow 0$, which is implied by the requirement of Proposition 2(ii). Take $\lambda = 2\sigma B_X \sqrt{2C_U} \{2 \log(2d/\alpha)/n\}^{1/2}$.

Then, with probability at least $1 - \alpha - \alpha' - \alpha''$ up to terms that tend to zero,

$$\left\| \hat{f} - f \right\|_n^2 \lesssim \frac{\sigma^2 B_X^2 C_U}{\kappa_1 c_U} \frac{s_0 \log(pqN_J)}{n} + \frac{\mathcal{B}_J^2}{\alpha'} + \frac{\sigma^2 p}{n\alpha''},$$

and, with no condition on the design, the same bound with $\lambda \|\boldsymbol{\theta}^*\|_1$ of Lemma S5.4 in place of the first term.

Proof. Intersect four events: $\mathcal{T}$, of probability at least $1 - \alpha$, by Lemma S8.2; $\{\hat{\sigma}_{\max}^2 \leq 2B_X^2 C_U\}$, by Lemma S5.3, which makes the stated λ admissible, that is $\lambda \geq 2\lambda_0$; $\{\lambda_{\min}(\hat{\Sigma}) \geq \gamma/2\}$, by Proposition 2(ii), which with (S5.2) gives $\tilde{\gamma} \geq \gamma/2$; and $\{\|\mathbf{b}\|_n^2 \leq \mathcal{B}_J^2/\alpha'\}$, which is Markov's inequality applied to $E\|\mathbf{b}\|_n^2 = \|f - f_J\|_{L_2(P)}^2 \leq \mathcal{B}_J^2$, the identity holding because the observations are identically distributed. Substituting in parts (ii) and (iii) of Theorem S8.1 gives the statement, and part (i) with Lemma S5.4 gives the unconditional version. $\square$

The first lemma is Theorem S8.1 read at an arbitrary comparator. It consumes no new step: the proof of that theorem uses the oracle only through the decomposition of $\mathbf{Y}$ and through the size of its support.

Lemma S8.3 (Theorem S8.1 with an arbitrary comparator). *Under Assumption 1, with $\mathbf{A}$ of rank p and $\lambda \geq 2\lambda_0$, on the event $\mathcal{T}$ the following hold simultaneously for every $\bar{\boldsymbol{\beta}} = (\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}})$, with $\bar{\mathbf{b}} = f - \mathbf{Z}\bar{\boldsymbol{\beta}}$, $\bar{M} = |\text{supp}(\bar{\boldsymbol{\theta}})|$ and $\bar{v} = \hat{\boldsymbol{\theta}} - \bar{\boldsymbol{\theta}}$: $\frac{1}{2} \left\| \tilde{\mathbf{B}}\bar{v} \right\|_n^2 + \lambda \left\| \hat{\boldsymbol{\theta}} \right\|_1 \leq 3\lambda \left\| \bar{\boldsymbol{\theta}} \right\|_1 + 2 \left\| \bar{\mathbf{b}} \right\|_n^2$; if $\tilde{\gamma} > 0$, $\left\| \tilde{\mathbf{B}}\bar{v} \right\|_n^2 \leq$*

$64\lambda^2\bar{M}/\tilde{\gamma} + 16 \|\bar{\mathbf{b}}\|_n^2$ and $\|\bar{v}\|_2^2 \leq 64\lambda^2\bar{M}/\tilde{\gamma}^2 + 16 \|\bar{\mathbf{b}}\|_n^2/\tilde{\gamma}$; and

$$\left\| \hat{f} - f \right\|_n^2 \leq 3 \min_{\bar{\beta}} \left\{ \frac{64\lambda^2\bar{M}}{\tilde{\gamma}} + 20 \|\bar{\mathbf{b}}\|_n^2 \right\} + 3 \|\Pi_{\mathbf{A}}\epsilon\|_n^2. \quad (\text{S8.7})$$

Proof. Follow the proof of Theorem S8.1 with $(\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}}, \bar{\mathbf{b}}, \text{supp}(\bar{\boldsymbol{\theta}}), \bar{M}, \bar{v})$ in place of $(\mathbf{c}, \boldsymbol{\theta}^*, \mathbf{b}, S_0, s_0, v)$. Step 1 uses only $\mathbf{Y} = \mathbf{A}\bar{\mathbf{c}} + \mathbf{B}\bar{\boldsymbol{\theta}} + \bar{\mathbf{b}} + \epsilon$, which is the definition of $\bar{\mathbf{b}}$, and $M_{\mathbf{A}}\mathbf{A} = 0$. Step 2 uses $\mathcal{T}$, which does not depend on the comparator, and $\|M_{\mathbf{A}}\bar{\mathbf{b}}\|_n \leq \|\bar{\mathbf{b}}\|_n$. Steps 3 to 5 are algebra with $\text{supp}(\bar{\boldsymbol{\theta}})$ in place of S_0 , using $\bar{\boldsymbol{\theta}}_{\text{supp}(\bar{\boldsymbol{\theta}})^c} = 0$, the Cauchy-Schwarz inequality and (S5.2). Step 6 gives $\hat{f} - \mathbf{Z}\bar{\beta} = \tilde{\mathbf{B}}\bar{v} + \Pi_{\mathbf{A}}(\bar{\mathbf{b}} + \epsilon)$ and hence $\hat{f} - f = \tilde{\mathbf{B}}\bar{v} + \Pi_{\mathbf{A}}(\bar{\mathbf{b}} + \epsilon) - \bar{\mathbf{b}}$. Since $\mathcal{T}$, $\tilde{\gamma}$ and the condition on λ do not involve $\bar{\beta}$, the inequalities hold simultaneously on the same event and the minimum may be taken; (S8.7) follows from $(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$. $\square$

Proof of Lemma S8.1

The envelope. Fix $\epsilon > 0$ and count $N(\epsilon) = \#\{a : |\theta_a^*| > \epsilon\}$. Inside a block and a level, Markov's inequality in the ℓ_π norm gives $\#\{k : |\theta_{\ell m, jk}^*| > \epsilon\} \leq \epsilon^{-\pi} \|\theta_{\ell m, j\cdot}^*\|_\pi^\pi$, and Assumption 5 gives $\|\theta_{\ell m, j\cdot}^*\|_\pi^\pi \leq C_g^\pi 2^{-j\pi(s+1/2-1/\pi)} = C_g^\pi 2^{j(1-a)}$ with $a = \pi(s+1/2) > 1$. Since a level has 2^j coordinates,

$$N_{\ell m}(\epsilon) \leq \sum_{j \geq 0} \min\{2^j, C_g^\pi \epsilon^{-\pi} 2^{j(1-a)}\}.$$

The two arguments agree at $2^{j_\epsilon a} = (C_g/\epsilon)^\pi$, that is at $2^{j_\epsilon} = (C_g/\epsilon)^\tau$, since $\pi/a = (s + 1/2)^{-1} = \tau$. Below that level the minimum is 2^j and the sum is at most 2^{j_ϵ} ; above it the minimum is the second term, which decreases geometrically with ratio $2^{1-a} < 1$, and the sum is $2^{j_\epsilon}/(1 - 2^{1-a})$. Adding the two parts and the pq blocks gives $N(\epsilon) \leq pqA_{s,\pi}(C_g/\epsilon)^\tau$; truncating at $j < J$ only decreases the count, so the bound holds for every J . Finally, $\epsilon < \theta_{(k)}^*$ implies $N(\epsilon) \geq k$, hence $\epsilon \leq C_g(pqA_{s,\pi}/k)^{1/\tau}$, and letting ϵ increase to $\theta_{(k)}^*$ gives the envelope; and $\tau = (s + 1/2)^{-1} < 2$ for every $s > 0$.

The tail. With $2/\tau > 1$, $\sum_{k>M} \theta_{(k)}^2 \leq C_\tau^2 \sum_{k>M} k^{-2/\tau} \leq C_\tau^2 \int_M^\infty x^{-2/\tau} dx = C_\tau^2 M^{1-2/\tau} \tau / (2 - \tau)$. □

Proof of Proposition S8.1

By parts (i) and (iii) of Theorem S8.1, on $\mathcal{T}$, $\|\tilde{\mathbf{B}}v\|_n^2 \leq 6\lambda \|\boldsymbol{\theta}^*\|_1 + 4\|\mathbf{b}\|_n^2$ and $\|\hat{f} - f\|_n^2 \leq 3(\|\tilde{\mathbf{B}}v\|_n^2 + 4\|\mathbf{b}\|_n^2 + \|\Pi_{\mathbf{A}}\boldsymbol{\epsilon}\|_n^2)$, and no property of $\hat{\Sigma}$ enters. Lemma S5.3 makes λ_n admissible with probability tending to one, since $2^{J_n} \log d_n/n \rightarrow 0$ in both regimes, and it is for this that the conditions $s' > 1/4$ in (i) and $s' > s/2$ in (ii) are needed: without them the choice in (i) gives $2^{J_n} \geq n^{1/(4s')} \geq n$, and the balance in (ii) gives $2^{J_n} \gtrsim n/\log n$; Markov's inequality applied to $\mathbb{E} \|\mathbf{b}\|_n^2 \leq \mathcal{B}_{J_n}^2$, from Lemma 2, and to $\mathbb{E}(\|\Pi_{\mathbf{A}}\boldsymbol{\epsilon}\|_n^2 \mid \mathbf{X}, \mathbf{U}) \leq \sigma^2 p/n$ gives the general statement. If

$s > 1/2$, Lemma S5.4 gives $\|\boldsymbol{\theta}^*\|_1 = O(1)$ and $\lambda_n \|\boldsymbol{\theta}^*\|_1 \asymp (J_n/n)^{1/2}$, and it suffices that the bias not dominate, that is $2^{-2J_n s'} \leq (J_n/n)^{1/2}$, which holds for $J_n \geq \log_2(n/J_n)/(4s')$. If $s < 1/2$, then $\|\boldsymbol{\theta}^*\|_1 \asymp 2^{J_n(1/2-s)}$, the two terms are $(J_n/n)^{1/2} 2^{J_n(1/2-s)}$ and $2^{-2J_n s'}$, and equating them gives $2^{J_n} \asymp (n/J_n)^{1/(1-2s+4s')}$ with common value $(J_n/n)^{2s'/(1-2s+4s')}$; when $s' = s$ the denominator is $1 + 2s$. $\square$

Proof of Theorem S8.2

By the definition of J_n , $(n/\log n)^{1/(2s'+1)} \leq 2^{J_n} < 2(n/\log n)^{1/(2s'+1)}$ and $J_n \asymp \log n$, so

$$\frac{pq2^{J_n} \log(pq2^{J_n})}{n} \asymp \frac{(n/\log n)^{1/(2s'+1)} \log n}{n} = \left(\frac{\log n}{n}\right)^{2s'/(2s'+1)} \rightarrow 0,$$

which is the requirement of Proposition 2(ii); (S5.2) then gives $\tilde{\gamma} \geq \gamma/2$ with probability tending to one. On that event Corollary S8.3 with $s_0 \leq d_n$ gives

$$\left\| \hat{f} - f \right\|_n^2 \lesssim \frac{\sigma^2 B_X^2 C_U}{\kappa_1 c_U} \frac{d_n L_n}{n} + \frac{\mathcal{B}_{J_n}^2}{\alpha'} + \frac{\sigma^2 p}{n\alpha''}.$$

With $d_n \asymp pq2^{J_n}$ and $L_n \asymp \log n$ the first term is the display above; the second is $\mathcal{B}_{J_n}^2 \asymp 2^{-2J_n s'} \leq (\log n/n)^{2s'/(2s'+1)}$ by the lower bound on 2^{J_n} ; and the third is $O(1/n)$, of strictly smaller order because $2s'/(2s'+1) < 1$. $\square$

Proof of Corollary S8.1

Inside a block the wavelets of level below J are orthonormal in $L_2[0, 1]$, so $\|\widehat{g}_{\ell m} - \Pi_J g_{\ell m}\|_{L_2[0,1]}^2 = \sum_{jk} v_{\ell m, jk}^2$, and summing over the pq blocks gives $\|v\|_2^2$; adding the bias of Lemma 2 in each block,

$$\sum_{\ell, m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 \leq 2 \|v\|_2^2 + \frac{2pqC_g^2}{1 - 2^{-2s'}} 2^{-2Js'}.$$

Theorem S8.1(ii) bounds $\|v\|_2^2$ by $64\lambda^2 s_0 / \widetilde{\gamma}^2 + 16 \|\mathbf{b}\|_n^2 / \widetilde{\gamma}$, and on the event of Theorem S8.2 the quantity $\widetilde{\gamma}$ is bounded below by the constant $\gamma/2$, so the two terms are of the same order as those of Theorem S8.2, the only difference being a constant with γ^{-2} in place of γ^{-1} . This is where the design condition is used a second time. For the levels, Lemma S5.1 gives $\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c}) = \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}) - \Pi_{\mathbf{A}}\mathbf{B}v$, so that $\|\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c})\|_n \leq \|\mathbf{b}\|_n + \|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n + \lambda_{\max}(\widehat{\Sigma})^{1/2} \|v\|_2$, while $\mathbf{A}^\top \mathbf{A}/n$ is a principal submatrix of $\widehat{\Sigma}$, whence $\|\widehat{\mathbf{c}} - \mathbf{c}\|_2^2 \leq \|\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c})\|_n^2 / (\gamma/2)$. Finally the components of a functional coefficient depend on distinct coordinates of u and the basis is orthonormal, so $\|\widehat{\beta}_\ell - \beta_\ell\|_{L_2([0,1]^q)} \leq |\widehat{c}_\ell - c_\ell| + \sum_m \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$, with q fixed. The same argument applies verbatim under the conditions of Theorem S8.3. □

Proof of Theorem S8.3

A nonasymptotic bound. Apply Lemma S8.3 with $\bar{\beta} = (\mathbf{c}, \boldsymbol{\theta}^{*,(M)})$, the best M term approximation of the oracle. Then $\bar{M} \leq M$ and $\bar{\mathbf{b}} = \mathbf{b} + \mathbf{B}(\boldsymbol{\theta}^* - \boldsymbol{\theta}^{*,(M)})$, so that, by the Cauchy-Schwarz inequality, Lemma S8.1 and (S6.2),

$$\|\bar{\mathbf{b}}\|_n^2 \leq 2\|\mathbf{b}\|_n^2 + 2\lambda_{\max}(\hat{\Sigma}) \left\| \boldsymbol{\theta}^* - \boldsymbol{\theta}^{*,(M)} \right\|_2^2 \leq 2\|\mathbf{b}\|_n^2 + \frac{2\Lambda C_\tau^2 \tau}{2-\tau} M^{1-2/\tau}.$$

Substituting in (S8.7), on the intersection of $\mathcal{T}$ with $\{\lambda_{\min}(\hat{\Sigma}) \geq \gamma/2\}$,

$$\left\| \hat{f} - f \right\|_n^2 \leq 3 \min_{1 \leq M \leq d} h(M) + 120 \|\mathbf{b}\|_n^2 + 3 \|\Pi_{\mathbf{A}} \boldsymbol{\epsilon}\|_n^2, \quad h(M) = \frac{64\lambda^2}{\tilde{\gamma}} M + \frac{40\Lambda C_\tau^2 \tau}{2-\tau} M^{1-2/\tau}.$$

The function h is strictly convex on the positive half line, since $1 - 2/\tau < 0$, and h' vanishes at $M^* = \{5\Lambda C_\tau^2 \tilde{\gamma} / (8\lambda^2)\}^{\tau/2}$, where $h(M^*) = 2(2 - \tau)^{-1}(64\lambda^2/\tilde{\gamma})M^*$; moreover $h(\lceil M^* \rceil) \leq h(M^*) + 64\lambda^2/\tilde{\gamma} \leq 2h(M^*)$ whenever $M^* \geq 1$, because $(2 - \tau)/2 < 1$.

The resolution. With $\lambda_n^2 \asymp \sigma^2 B_X^2 C_U J_n / n$, $\tilde{\gamma} \in [\gamma/2, \Lambda]$ and the other constants fixed, $M_n^* \asymp \lambda_n^{-\tau} \asymp (n/J_n)^{\tau/2}$ and $h(M_n^*) \asymp \lambda_n^2 M_n^* \asymp (J_n/n)^{(2-\tau)/2}$. Writing $J_n = \lceil c \log_2 n \rceil$, so that $2^{J_n} \asymp n^c$, the three inequalities of the window (S8.3) are, in order: $M_n^* \leq d_n$, which asks $(n/J_n)^{\tau/2} \lesssim pq n^c$, that is $c \geq \tau/2$ up to logarithms; that the bias not dominate, $n^{-2cs'} \lesssim (J_n/n)^{(2-\tau)/2}$, that is $c \geq (2 - \tau)/(4s')$; and the requirement of Proposition 2(ii), $n^{c-1} \log n \rightarrow 0$, that is $c < 1$. The term $\sigma^2 p/n$ is of smaller order because $(2 - \tau)/2 < 1$. The window is nonempty if and only if $\tau/2 < 1$,

which always holds, and $(2 - \tau)/(4s') < 1$, that is $s' > (2 - \tau)/4$.

Under Assumption 5 the hypothesis is automatic. By Lemma S8.1 the envelope holds with $\tau = (s+1/2)^{-1}$, whence $2 - \tau = 4s/(2s+1)$, $(2 - \tau)/2 = 2s/(2s+1)$ and $(2 - \tau)/4 = s/(2s+1)$, which is the condition $s' > s/(2s+1)$ of the statement; for $\pi \geq 2$ it holds for every $s > 0$, and for $\pi < 2$ it reads $2s^2/(2s+1) > 1/\pi - 1/2$. The rate for the components is the argument of Corollary S8.1 with $\left\| \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \right\|_2 \leq \|\bar{v}\|_2 + \left\| \boldsymbol{\theta}^* - \boldsymbol{\theta}^{*,(M)} \right\|_2$ and Lemma S8.3, the only difference being an extra factor $\tilde{\gamma}^{-1}$ in the first term, which does not change the order. Finally $2s/(2s+1) \geq 2s'/(2s'+1)$, with equality only when $s' = s$, by monotonicity of $x \mapsto 2x/(2x+1)$ and $s' \leq s$, so the rate is never worse than that of Theorem S8.2 and is strictly better when $\pi < 2$. □

Remark S8.2 (What compressibility buys). Read at the oracle, Theorem S8.1(ii) has $s_0 = d$ and delivers the rate of linear approximation; read at a sparse comparator, through Lemma S8.3, it delivers the rate of Theorem S8.3. It is the same inequality in both cases, and neither says anything about $\text{supp}(\widehat{\boldsymbol{\theta}})$: no minimal signal and no irrepresentability condition is used, and what is gained is a rate governed by the effective dimension M_n^* rather than by d_n .

Proof of Corollary S8.2

A *nonasymptotic bound on Δ* . Work at $J = J_n$ and $\lambda = \lambda_n$, and intersect the four events of the proof of Corollary S8.3: $\mathcal{T}$, of probability at least $1 - \alpha$, by Lemma S8.2; $\{\hat{\sigma}_{\max}^2 \leq 2B_X^2 C_U\}$, by Lemma S5.3, which makes λ_n admissible; $\{\lambda_{\min}(\hat{\Sigma}) \geq \gamma/2\}$, by Proposition 2(ii), which with (S5.2) gives $\tilde{\gamma} \geq \gamma/2$; and $\{\|\mathbf{b}\|_n^2 \leq \mathcal{B}_{J_n}^2/\alpha'\}$, by Markov's inequality and Lemma 2. On this event Theorem S8.1(ii) with $s_0 \leq d_n$ gives

$$\|v\|_2^2 \leq \frac{64\lambda_n^2 d_n}{\tilde{\gamma}^2} + \frac{16\|\mathbf{b}\|_n^2}{\tilde{\gamma}} \leq \frac{256\lambda_n^2 d_n}{\gamma^2} + \frac{32\mathcal{B}_{J_n}^2}{\alpha'\gamma}.$$

For a fixed block, the orthonormality of the wavelets of level below J_n gives $\|\hat{g}_{\ell m} - \Pi_J g_{\ell m}\|_{L_2[0,1]} = \|v_{\ell m,\cdot}\|_2 \leq \|v\|_2$, and the first display of Lemma 2 bounds $\|\Pi_J g_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2$ by $C_g^2(1 - 2^{-2s'})^{-1}2^{-2J_n s'}$. By $(x + y)^2 \leq 2x^2 + 2y^2$,

$$\Delta^2 \leq \bar{\Delta}_n^2 = \frac{512\lambda_n^2 d_n}{\gamma^2} + \frac{64\mathcal{B}_{J_n}^2}{\alpha'\gamma} + \frac{2C_g^2}{1 - 2^{-2s'}} 2^{-2J_n s'} \quad (\text{S8.8})$$

with probability at least $1 - \alpha - \alpha' - \mathbb{P}\{\lambda_{\min}(\hat{\Sigma}) < \gamma/2\} - \mathbb{P}\{\hat{\sigma}_{\max}^2 > 2B_X^2 C_U\}$, the last two terms tending to zero by Proposition 2(ii) and Lemma S5.3; their sum is the ω_n of Corollary S8.2(iii), and this paragraph is the proof of that part. By Lemma S7.1(iii), on that event $\hat{\mathcal{S}}(t) = \mathcal{S}$ for every t with $\bar{\Delta}_n \leq t < \nu_{\min} - \bar{\Delta}_n$, a window that is nonempty whenever $\nu_{\min} > 2\bar{\Delta}_n$. The bound $\bar{\Delta}_n$ depends on the components only through the constants of

Assumption 5, and the two residual probabilities do not depend on them at all, so everything above holds uniformly over the class, which is what allows the target to vary with n in Assumption 6.

The order of $\bar{\Delta}_n$. At the resolution of Theorem S8.2 the three terms of (S8.8) are at most of order ρ_n^2 : the first because $\lambda_n^2 d_n \asymp pq2^{J_n} J_n/n$, which is the quantity shown to be of order $(\log n/n)^{2s'/(2s'+1)}$ in the proof of Theorem S8.2, and the other two because $\mathcal{B}_{J_n}^2 \asymp 2^{-2J_n s'} \leq (\log n/n)^{2s'/(2s'+1)}$ by the definition of J_n . Hence $\bar{\Delta}_n \leq c(\alpha, \alpha')\rho_n$ with $c(\alpha, \alpha') < \infty$. Given $\epsilon > 0$, take $\alpha = \alpha' = \epsilon/4$ and n large enough that the two residual probabilities add up to less than $\epsilon/2$; then $P\{\Delta > c(\epsilon/4, \epsilon/4)\rho_n\} < \epsilon$, that is $\Delta = O_p(\rho_n)$.

Part (i). If $t_n/\rho_n \rightarrow \infty$, then $P(\Delta \leq t_n) \rightarrow 1$, and on that event Lemma S7.1(i) with $\bar{\Delta} = \Delta$ gives $\widehat{\mathcal{S}}(t_n) \subseteq \mathcal{S}$. No property of $\nu_{\min, n}$ has been used.

Part (ii). If $\mathcal{S}$ is empty, part (ii) is part (i). Otherwise, the condition $\limsup_n t_n/\nu_{\min, n} < 1$ gives $\eta > 0$ and n_0 with $\nu_{\min, n} - t_n \geq \eta \nu_{\min, n}$ for $n \geq n_0$, and Assumption 6 with ρ_n in place of $\rho_n^{\mathcal{G}}$ gives $(\nu_{\min, n} - t_n)/\rho_n \geq \eta \nu_{\min, n}/\rho_n \rightarrow \infty$. Hence $P(\Delta \leq t_n) \rightarrow 1$ and $P(\Delta < \nu_{\min, n} - t_n) \rightarrow 1$, and on the intersection of the two events Lemma S7.1(iii) with $\bar{\Delta} = \Delta$ gives $\widehat{\mathcal{S}}(t_n) = \mathcal{S}$. For the existence, $t_n = (\rho_n \nu_{\min, n})^{1/2}$ has $t_n/\rho_n = (\nu_{\min, n}/\rho_n)^{1/2} \rightarrow \infty$ and $t_n/\nu_{\min, n} = (\rho_n/\nu_{\min, n})^{1/2} \rightarrow 0$. $\square$

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At the resolution of Theorem S8.3. Theorem S8.3 gives $\sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 = O_p(\tilde{\rho}_n^2)$ with $\tilde{\rho}_n = (\log n/n)^{s/(2s+1)}$, hence $\Delta = O_p(\tilde{\rho}_n)$, and the bound in its proof depends on the components only through the constants of Assumption 5, through C_τ of Lemma S8.1 and the design constants, so that this holds uniformly over the class. The proofs of parts (i) and (ii) above use nothing about Δ beyond $\Delta = O_p(\rho_n)$ uniformly over the class, together with Lemma S7.1 and Assumption 6, and they go through verbatim with $\tilde{\rho}_n$ in place of ρ_n .

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